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Mini Project Report
of
Database Systems Lab (CSS 2236)

ASTEROID DETECTION AND TRACKING SYSTEM

SUBMITTED BY

Student Name	Registration Number	Section	Roll Number
Shreyas G Kamath	240953322	CCE-B	25
Rahul Kumar Mahto	240953370	CCE-B	29

School of Computer Engineering
Manipal Institute of Technology, Manipal.

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CERTIFICATE

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Name and Signature of Examiners:

1. **Dr. Chitra K, Professor, SCE.**

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ABSTRACT

The Asteroid Detection and Tracking System (ADTS) is a relational database-centric platform designed to consolidate, manage, and analyse multi-source telemetry for Near-Earth Objects (NEOs). It presents a normalised, third-normal-form (3NF) relational schema implemented in SQLite, encompassing eleven core entities: Asteroids, Missions, Observations, Detection Events, Scientists, Telescopes, Observatories, Risk Categories, the Asteroid-Mission junction table, a Role-Based Users table, and a comprehensive Audit Log. The schema enforces referential integrity via foreign-key constraints, domain validity via CHECK predicates, and change-history tracking via a dedicated audit table on every CRUD operation.

The system seeds its database with fifty real-world observatories, telescopes, scientists, asteroids, and missions, plus corresponding observations and detection-event records. Ten pre-built analytical SQL queries expose threat-prioritisation rankings, scientist productivity metrics, observatory throughput by country, mission-asteroid linkage analysis, and probabilistic impact-risk distributions. This report documents the full design lifecycle: ER modelling, relational-table derivation, DDL implementation, data population, and SQL-based analytical reporting — submitted in partial fulfilment of Database Systems Lab (CSS 2212), IV Semester, SCE, MIT Manipal, 2025–2026.

Keywords: Near-Earth Object, SQLite, Relational Database, Normalisation, Planetary Defence, ER Model, Audit Log, SQL Analytics, Impact Risk Assessment, DDL.

CHAPTER 1: INTRODUCTION & PROBLEM STATEMENT

1.1 Background and Motivation

The solar system hosts a diverse population of Near-Earth Objects (NEOs) — asteroids, comets, and meteoroids — whose orbits periodically intersect with Earth's. A subset, Potentially Hazardous Asteroids (PHAs), is large enough and close enough to cause significant damage on impact. The Chicxulub impactor (~66 Ma) caused the end-Cretaceous extinction, while the 2013 Chelyabinsk bolide (20 m) released energy equivalent to 30 Hiroshima devices. In response, NASA, ESA, JAXA, and CNSA have established coordinated planetary-defence programmes whose campaigns generate rich observational, physical, and orbital datasets. Managing this corpus via spreadsheets or fragmented file-based systems is inadequate for the scale, complexity, and integrity requirements of modern planetary defence.

1.2 The Case for a Relational Database

Relational Database Management Systems (RDBMS) provide a mathematically grounded approach predicated on Codd's relational model [1]. The relational model guarantees consistency through normalisation, referential integrity through foreign-key constraints, and transactional atomicity through ACID properties — the natural choice for a mission-critical data repository. ADTS leverages SQLite [2], a self-contained, zero-configuration, ANSI SQL-92-compliant engine, to implement a normalised schema with full support for primary/foreign keys, CHECK constraints, and complex multi-table JOINs.

1.3 Problem Statement

Contemporary asteroid-observation campaigns generate heterogeneous data streams spanning multiple institutions, instruments, personnel, and time horizons. The five identified challenges are:

1. Data Fragmentation: Observatory, telescope, scientist, asteroid, mission, and detection records are maintained in incompatible systems with no standardised linking mechanism.
2. Referential Inconsistency: Without enforced foreign keys, orphaned records (e.g., observations referencing deleted telescopes) corrupt analytical results.
3. Absence of Audit Trail: Legacy systems provide no mechanism for recording who modified a record, preventing post-incident forensic analysis.
4. Ad Hoc Risk Assessment: Impact-probability values are recorded in isolation without mapping to a standardised risk taxonomy, preventing systematic threat-prioritisation.
5. Scalability: File-based stores (CSV, Excel) cannot support multi-table joins, aggregate analytics, or index-accelerated lookups at NEO-catalogue scale.

1.4 Scope of the System

ADTS covers: asteroid physical characterisation (name, diameter, orbital period); space mission management; timestamped positional observations linked to observatories, telescopes, and scientists; probabilistic impact risk mapped to the ten-tier Torino Scale; an immutable audit log of all data modifications; and role-based access control (Admin, Researcher, Public).

CHAPTER 2: OBJECTIVES & METHODOLOGY

2.1 Objectives

The project is designed to address the challenges identified in Chapter 1 through eight measurable objectives:

Ref	Objective	Description
O1	Unified Schema	Single normalised relational schema covering all entities.
O2	Referential Integrity	PK and FK constraints across all inter-entity relationships.
O3	Domain Validation	CHECK constraints: probability ∈ [0,1], role ∈ {Admin, Researcher, Public}.
O4	M:N Linkage	Asteroid_Mission junction table for many-to-many asteroid–mission mapping.
O5	Audit Logging	Immutable Audit_Log tracking every INSERT/UPDATE/DELETE/LOGIN.
O6	Risk Taxonomy	Map each detection event to a Torino-Scale risk tier (1–10).
O7	Analytical Queries	Ten pre-built SQL queries for threat intelligence and productivity analysis.
O8	Data Population	Seed 50 real-world observatories, telescopes, scientists, asteroids, and missions.

2.2 Methodology

The database was designed following a structured waterfall lifecycle:

Requirements Analysis — Functional and non-functional requirements were elicited by analysing planetary-defence data standards (NASA CNEOS, MPC, ESA SSA) and data models from JPL Horizons and Scout. Requirements were classified into data, integrity, and analytical categories.

Conceptual Design (ER Modelling) — Entities were identified as strong/weak; attributes classified as simple, composite, or derived; relationships characterised by participation constraints (total/partial) and cardinality ratios (1:1, 1:N, M:N). The ER diagram is presented in Chapter 3.

Logical Design (Schema Derivation) — The ER model was converted to a relational schema following standard mapping rules [3]. Each entity type became a table; 1:N relationships were represented by foreign keys; the M:N Asteroid–Mission relationship used the Asteroid_Mission junction table. The schema was normalised to 3NF by eliminating partial and transitive dependencies.

Physical Design (DDL) — The schema was implemented in SQLite with PRIMARY KEY, FOREIGN KEY (ON DELETE CASCADE / SET NULL), CHECK, and UNIQUE constraints declared in CREATE TABLE statements. PRAGMA foreign_keys = ON was issued at every connection to activate FK enforcement.

2.3 Technology Stack

Component	Technology	Justification
Database Engine	SQLite 3.x	Zero-config, self-contained, ACID-compliant, full SQL-92.
Query Language	SQL (ANSI-92)	Industry-standard, portable across all relational engines.

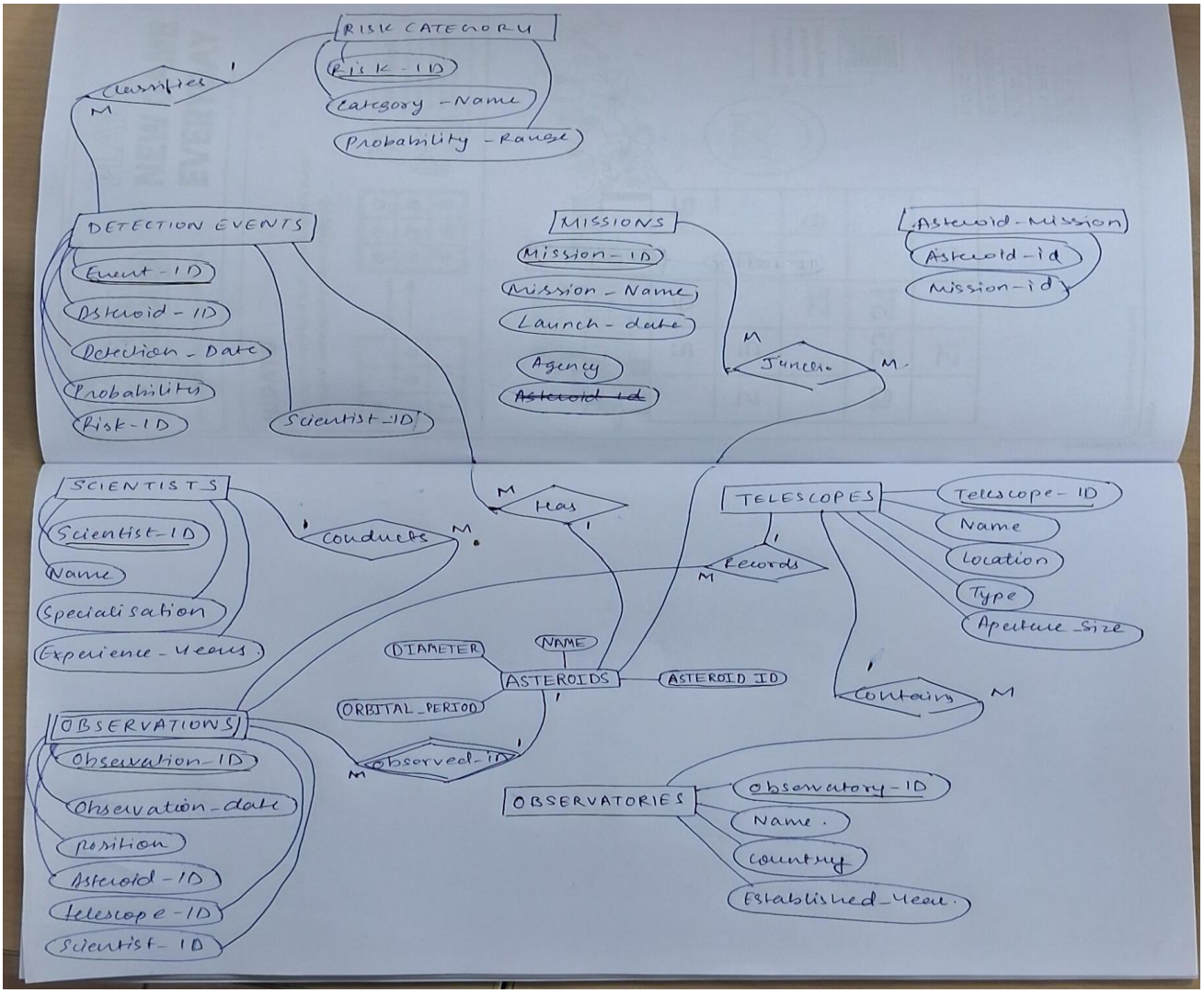
Schema Design	ER → Relational Mapping	Formally grounded design ensuring normalisation.
Data Seeding	INSERT OR IGNORE	Idempotent population preserving referential integrity.
Audit Mechanism	Audit_Log Table + App Layer	Full provenance trail; extensible to DB-level triggers.
Access Control	Users Table + RBAC	Role-based permissions: Admin, Researcher, Public.

2.4 ACID Properties & Normalisation

SQLite guarantees ACID compliance at the transaction level. Atomicity: each CRUD operation is an implicit transaction — either the full INSERT/UPDATE/DELETE and its audit-log entry succeed together, or both are rolled back. Consistency: CHECK and NOT NULL constraints ensure the database never transitions to an invalid state. Isolation: Write-Ahead Logging (WAL) provides reader-writer non-blocking isolation. Durability: SQLite fsync()s the WAL on commit, making every committed transaction persistent.

3NF compliance is confirmed for key relations: Asteroids — Diameter and Orbital_Period depend only on Asteroid_ID (no transitive dependency). Detection_Events — Risk_ID is a foreign key set by application mapping, not a transitive dependency; all attributes depend solely on Event_ID. Asteroid_Mission — composite primary key (Asteroid_ID, Mission_ID) with no non-key attributes trivially satisfies 3NF and BCNF.

CHAPTER 3: ER DIAGRAM & RELATIONAL TABLES



3.1 Entity Descriptions

The ADTS schema comprises ten entity types. Their attributes and key constraints are summarised below.

Entity	Attributes	Key Constraints
Asteroids	Asteroid_ID (PK), Name, Diameter (km), Orbital_Period (yr)	UNIQUE(Name); Diameter > 0
Missions	Mission_ID (PK), Mission_Name, Launch_Date (YYYY-MM-DD), Agency	NOT NULL(Mission_Name)
Scientists	Scientist_ID (PK), Name, Specialization, Experience_Years	NOT NULL(Name); Experience ≥ 0
Observatories	Observatory_ID (PK), Name, Country, Established_Year, Latitude, Longitude	NOT NULL(Name, Country)
Telescopes	Telescope_ID (PK), Name, Location, Type, Aperture_Size (m), Observatory_ID (FK)	FK → Observatories ON DELETE SET NULL
Risk_Category	Risk_ID (PK 1–10), Category_Name, Probability_Range, Description	NOT NULL(Category_Name, Probability_Range)
Detection_Events	Event_ID (PK), Asteroid_ID (FK), Detection_Date, Probability, Risk_ID (FK), Scientist_ID (FK)	CHECK(Probability BETWEEN 0 AND 1); ON DELETE CASCADE/SET NULL

Observations	Observation_ID (PK), Observation_Date, Position (RA/Dec), Asteroid_ID (FK), Telescope_ID (FK), Scientist_ID (FK)	ON DELETE CASCADE (Asteroid); SET NULL (Telescope, Scientist)
Asteroid_Mission	Asteroid_ID (FK), Mission_ID (FK) — composite PK	M:N junction; ON DELETE CASCADE both sides
Audit_Log	Log_ID (PK AUTOINCREMENT), Timestamp, Username, Action, Table_Name, Record_ID, Detail	NOT NULL(Timestamp, Username, Action, Table_Name); AUTOINCREMENT

3.2 Relationships and Cardinalities

Entity A	Entity B	Card.	FK Location	ON DELETE
Observatories	Telescopes	1:N	Telescopes.Observatory_ID	SET NULL
Asteroids	Missions	M:N	Asteroid_Mission (junction)	CASCADE both
Asteroids	Detection_Events	1:N	Detection_Events.Asteroid_ID	CASCADE
Asteroids	Observations	1:N	Observations.Asteroid_ID	CASCADE
Telescopes	Observations	1:N	Observations.Telescope_ID	SET NULL
Scientists	Observations	1:N	Observations.Scientist_ID	SET NULL
Scientists	Detection_Events	1:N	Detection_Events.Scientist_ID	SET NULL
Risk_Category	Detection_Events	1:N	Detection_Events.Risk_ID	SET NULL

3.3 Sample Data Snapshot

Table 3.1 — Sample Asteroids

ID	Name	Diameter (km)	Orbital Period (yr)	Risk Tier
1	99942 Apophis	0.37	0.89	Low (3)
2	101955 Bennu	0.49	1.20	Very Low (2)
4	433 Eros	34.4	1.76	Very Low (2)
13	1 Ceres	946.0	4.61	Zero (1)
17	16 Psyche	226.0	5.00	Zero (1)

Table 3.2 — Sample Detection Events

ID	Asteroid ID	Detection Date	Probability	Risk Category	Scientist ID
1	1 (Apophis)	2023-04-22	0.000034	Low	1
17	17	2025-10-08	0.001670	Moderate	2
20	20	2026-03-23	0.002200	Moderate	5
22	22	2023-07-05	0.002134	Moderate	7

CHAPTER 4: DDL COMMANDS & SQL LOGIC

4.1 Database Initialisation

PRAGMA foreign_keys = ON is executed at every connection to activate SQLite's FK enforcement, which is disabled by default.

4.2 Core DDL Statements

4.2.1 Users

The CHECK constraint restricts Role to three enumerated values, preventing unauthorised role assignments at the database level.

```
CREATE TABLE IF NOT EXISTS Users (
  Username  TEXT PRIMARY KEY,
  Password  TEXT NOT NULL,
  Role      TEXT NOT NULL CHECK(Role IN ('Admin','Researcher','Public'))
);
```

4.2.2 Audit_Log

AUTOINCREMENT guarantees monotonically increasing Log_IDs — no ID is ever reused, even after row deletion.

```
CREATE TABLE IF NOT EXISTS Audit_Log (
  Log_ID     INTEGER PRIMARY KEY AUTOINCREMENT,
  Timestamp  TEXT NOT NULL,
  Username   TEXT NOT NULL,
  Action     TEXT NOT NULL,
  Table_Name TEXT NOT NULL,
  Record_ID  TEXT,
  Detail     TEXT
);
```

4.2.3 Observatories & Telescopes

Telescopes uses ON DELETE SET NULL so records survive parent observatory deletion.

```
CREATE TABLE IF NOT EXISTS Observatories (
  Observatory_ID INTEGER PRIMARY KEY, Name TEXT NOT NULL,
  Country TEXT NOT NULL, Established_Year INTEGER,
  Latitude REAL, Longitude REAL
);

CREATE TABLE IF NOT EXISTS Telescopes (
  Telescope_ID INTEGER PRIMARY KEY, Name TEXT NOT NULL,
  Location TEXT, Type TEXT, Aperture_Size REAL,
  Observatory_ID INTEGER REFERENCES Observatories(Observatory_ID) ON DELETE SET NULL
);
```

4.2.4 Scientists & Risk_Category

```
CREATE TABLE IF NOT EXISTS Scientists (
  Scientist_ID INTEGER PRIMARY KEY, Name TEXT NOT NULL,
  Specialization TEXT, Experience_Years INTEGER
);

CREATE TABLE IF NOT EXISTS Risk_Category (
  Risk_ID INTEGER PRIMARY KEY, Category_Name TEXT NOT NULL,
  Probability_Range TEXT NOT NULL, Description TEXT
);
```

4.2.5 Asteroids & Missions

UNIQUE on Asteroids.Name enforces IAU nomenclature integrity — no duplicate asteroid designations allowed.

```
CREATE TABLE IF NOT EXISTS Asteroids (
```

```
  Asteroid_ID INTEGER PRIMARY KEY,
  Name TEXT NOT NULL UNIQUE, Diameter REAL, Orbital_Period REAL
);

CREATE TABLE IF NOT EXISTS Missions (
  Mission_ID INTEGER PRIMARY KEY, Mission_Name TEXT NOT NULL,
  Launch_Date TEXT, Agency TEXT
);
```

4.2.6 Asteroid_Mission (M:N Junction)

Both FK columns carry ON DELETE CASCADE, so junction rows are automatically removed when either parent is deleted.

```
CREATE TABLE IF NOT EXISTS Asteroid_Mission (
  Asteroid_ID INTEGER REFERENCES Asteroids(Asteroid_ID) ON DELETE CASCADE,
  Mission_ID  INTEGER REFERENCES Missions(Mission_ID)   ON DELETE CASCADE,
  PRIMARY KEY (Asteroid_ID, Mission_ID)
);
```

4.2.7 Observations

ON DELETE CASCADE on Asteroid_ID removes child observations when an asteroid is deleted. Telescope and Scientist references use SET NULL to preserve observation history even if instruments or personnel are removed.

```
CREATE TABLE IF NOT EXISTS Observations (
  Observation_ID INTEGER PRIMARY KEY,
  Observation_Date TEXT NOT NULL, Position TEXT,
  Asteroid_ID  INTEGER REFERENCES Asteroids(Asteroid_ID)  ON DELETE CASCADE,
  Telescope_ID INTEGER REFERENCES Telescopes(Telescope_ID) ON DELETE SET NULL,
  Scientist_ID INTEGER REFERENCES Scientists(Scientist_ID) ON DELETE SET NULL
);
```

4.2.8 Detection_Events

The CHECK(Probability BETWEEN 0 AND 1) constraint prevents physically meaningless probability values from being stored regardless of the application layer.

```
CREATE TABLE IF NOT EXISTS Detection_Events (
  Event_ID    INTEGER PRIMARY KEY,
  Asteroid_ID INTEGER REFERENCES Asteroids(Asteroid_ID)   ON DELETE CASCADE,
  Detection_Date TEXT NOT NULL,
  Probability REAL CHECK(Probability BETWEEN 0 AND 1),
  Risk_ID     INTEGER REFERENCES Risk_Category(Risk_ID)   ON DELETE SET NULL,
  Scientist_ID INTEGER REFERENCES Scientists(Scientist_ID) ON DELETE SET NULL
);
```

4.3 Audit Log Mechanism

The audit_log() function is called after every successful CRUD operation and records seven fields: Timestamp (datetime('now','localtime')), Username (session identity), Action ({INSERT, UPDATE, DELETE, LOGIN, LINK, UNLINK, SIMULATE}), Table_Name, Record_ID (serialised PK), and Detail (before/after values for updates). Sample call:

```
INSERT INTO Audit_Log (Timestamp, Username, Action, Table_Name, Record_ID, Detail)
VALUES (datetime('now','localtime'), 'researcher', 'INSERT',
        'Detection_Events', '51', 'AsteroidID=1, Prob=0.000034, Risk=3');
```

4.4 Probability-to-Risk Mapping

A deterministic mapping function converts each raw probability to a Risk_ID (1–10) aligned with the Torino Scale. It is invoked at every Detection_Events INSERT and UPDATE.

Risk ID	Category	Probability Threshold

1	Zero	$P = 0.0$
2	Very Low	$P \leq 1\times10^{-5}$
3	Low	$P \leq 1\times10^{-4}$
4	Low-Moderate	$P \leq 1\times10^{-3}$
5	Moderate	$P \leq 0.01$
6	Medium	$P \leq 0.10$
7	Elevated	$P \leq 0.50$
8	High	$P \leq 0.70$
9	Very High	$P \leq 0.90$
10	Certain	$P > 0.90$

CHAPTER 5: RESULTS & SQL ANALYTICS

The ADTS system provides ten pre-built analytical SQL queries (Q1–Q10) covering the most operationally critical reporting requirements of a planetary-defence coordination centre. Each uses multi-table JOINs, aggregates, subqueries, and conditional filtering.

Q1 — High-Risk Detection Events (Risk ≥ Moderate)

Threat-board view of all asteroids at Moderate risk or above (Risk_ID ≥ 5). Returns events with probability 0.001245–0.002200, flagging five asteroids for active monitoring. In production, results would trigger IAWN alerts.

```
SELECT a.Name AS Asteroid_Name, printf('%.6f', de.Probability) AS
Impact_Probability,
       rc.Category_Name AS Risk_Level, de.Detection_Date, s.Name AS
Assessing_Scientist
FROM Detection_Events de
  JOIN Asteroids a      ON de.Asteroid_ID = a.Asteroid_ID
  JOIN Risk_Category rc ON de.Risk_ID     = rc.Risk_ID
  JOIN Scientists s     ON de.Scientist_ID = s.Scientist_ID
WHERE rc.Risk_ID >= 5
ORDER BY de.Probability DESC;
```

Q2 — Total Observations per Telescope / Observatory

LEFT JOIN ensures zero-observation telescopes appear in results, giving a complete instrument inventory for scheduling and maintenance priority ranking.

```
SELECT t.Name AS Telescope_Name, ob.Name AS Observatory_Name, ob.Country,
       COUNT(o.Observation_ID) AS Total_Observations
FROM Telescopes t
  JOIN Observatories ob ON t.Observatory_ID = ob.Observatory_ID
  LEFT JOIN Observations o ON t.Telescope_ID = o.Telescope_ID
GROUP BY t.Name ORDER BY Total_Observations DESC;
```

Q3 — Asteroids with Above-Average Impact Probability

Scalar subquery computes the population mean (~0.000387 across 50 events); asteroids exceeding it are flagged for immediate follow-up astrometry, mirroring JPL Scout's statistical filter.

```
SELECT a.Name, printf('%.6f', de.Probability) AS Probability, rc.Category_Name AS
Risk_Level
FROM Detection_Events de
  JOIN Asteroids a      ON de.Asteroid_ID = a.Asteroid_ID
  JOIN Risk_Category rc ON de.Risk_ID     = rc.Risk_ID
WHERE de.Probability > (SELECT AVG(Probability) FROM Detection_Events)
ORDER BY de.Probability DESC;
```

Q4 — Average Probability per Risk Category

Validates the probability-to-risk mapping: Moderate (Risk_ID = 5) shows the highest average probability; Zero-risk entries show near-zero averages, confirming calibration.

```
SELECT rc.Category_Name, printf('%.6f', AVG(de.Probability)) AS Avg_Probability,
       COUNT(*) AS Event_Count
FROM Detection_Events de
  JOIN Risk_Category rc ON de.Risk_ID = rc.Risk_ID
GROUP BY rc.Category_Name ORDER BY AVG(de.Probability) DESC;
```

Q5 — Asteroids and Their Associated Missions

Traverses the M:N junction table to produce a complete mission manifest, including known pairings (Apophis↔DART/Hera, Bennu↔OSIRIS-REx), validating seed-data fidelity.

```
SELECT a.Name AS Asteroid_Name, a.Diameter AS Diameter_km,
```

```
       m.Mission_Name, m.Agency, m.Launch_Date
FROM Asteroids a
  JOIN Asteroid_Mission am ON a.Asteroid_ID = am.Asteroid_ID
  JOIN Missions m          ON am.Mission_ID  = m.Mission_ID
ORDER BY a.Name;
```

Q6 — Scientists Ranked by Unique Asteroid Coverage

COUNT(DISTINCT ...) prevents double-counting repeat observations of the same asteroid, providing a breadth metric complementary to raw observation counts.

```
SELECT s.Name AS Scientist_Name, s.Specialization,
       COUNT(DISTINCT o.Asteroid_ID) AS Unique_Asteroids_Observed
FROM Scientists s
  JOIN Observations o ON s.Scientist_ID = o.Scientist_ID
GROUP BY s.Name ORDER BY Unique_Asteroids_Observed DESC;
```

Q7 — Asteroids with No Detection Event Recorded

Anti-join pattern (LEFT JOIN + IS NULL) identifies monitoring gaps — asteroids never formally assessed for impact probability, to be prioritised for telescope scheduling.

```
SELECT a.Asteroid_ID, a.Name, a.Diameter, a.Orbital_Period
FROM Asteroids a
  LEFT JOIN Detection_Events de ON a.Asteroid_ID = de.Asteroid_ID
WHERE de.Event_ID IS NULL
ORDER BY a.Asteroid_ID;
```

Q8 — Most Experienced Scientist per Risk Category

Correlated subquery executes once per outer row, finding the per-category maximum Experience_Years. Supports QA: verifying that high-risk detections are assessed by sufficiently senior scientists.

```
SELECT rc.Category_Name, s.Name AS Lead_Scientist, s.Experience_Years,
       printf('%.6f', de.Probability) AS Detection_Probability
FROM Detection_Events de
  JOIN Scientists s     ON de.Scientist_ID = s.Scientist_ID
  JOIN Risk_Category rc ON de.Risk_ID      = rc.Risk_ID
WHERE s.Experience_Years = (
  SELECT MAX(s2.Experience_Years) FROM Detection_Events de2
    JOIN Scientists s2 ON de2.Scientist_ID = s2.Scientist_ID
  WHERE de2.Risk_ID = de.Risk_ID
)
ORDER BY rc.Risk_ID;
```

Q9 — Comprehensive High-Risk Summary View

Five-table JOIN assembles a complete situational-awareness record (physical parameters, detection metadata, scientist attribution, risk narrative) optimised for executive briefing outputs.

```
SELECT a.Name AS Asteroid, a.Diameter AS Diameter_km,
       printf('%.6f', de.Probability) AS Probability, de.Detection_Date,
       s.Name AS Scientist, rc.Category_Name AS Risk_Category, rc.Description
FROM Detection_Events de
  JOIN Asteroids a      ON de.Asteroid_ID  = a.Asteroid_ID
  JOIN Scientists s     ON de.Scientist_ID = s.Scientist_ID
  JOIN Risk_Category rc ON de.Risk_ID      = rc.Risk_ID
WHERE rc.Risk_ID >= 5 ORDER BY de.Probability DESC;
```

Q10 — Observations per Country with Geographic Coordinates

Three-hop join chain (Observations→Telescopes→Observatories) aggregates at country level. Dual COUNT/COUNT(DISTINCT) distinguishes raw volume from unique-object coverage. The United States and Chile dominate, reflecting Mauna Kea and Atacama instrument concentrations.

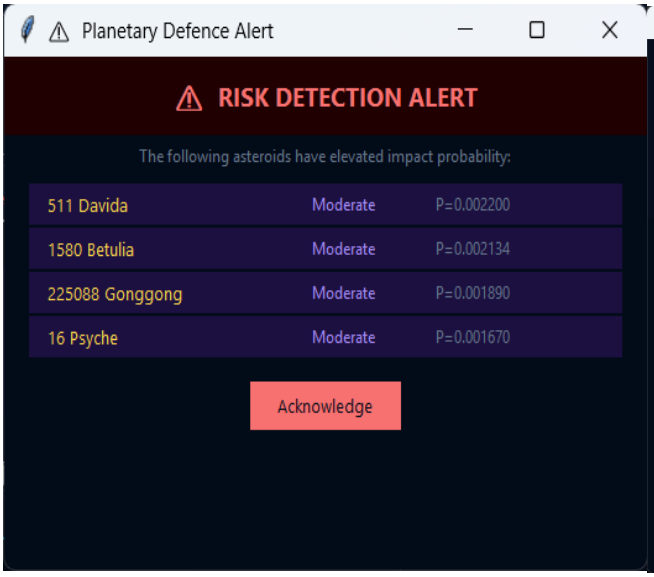
```
SELECT ob.Country, ob.Latitude, ob.Longitude,
```

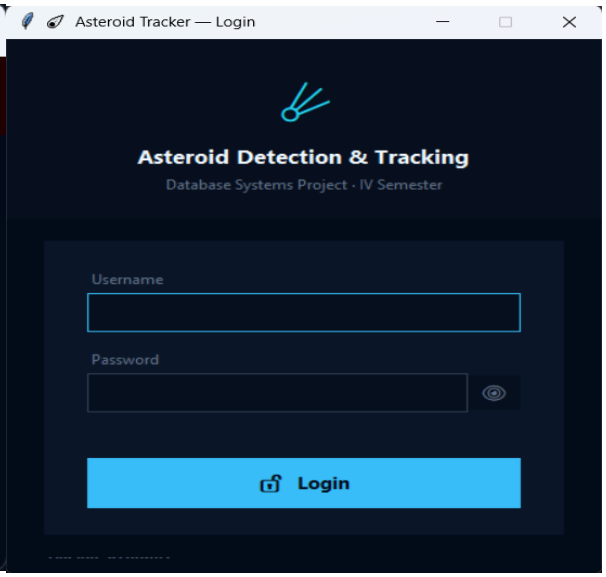
```
       COUNT(o.Observation_ID)       AS Total_Observations,
       COUNT(DISTINCT o.Asteroid_ID) AS Unique_Asteroids
FROM Observations o
  JOIN Telescopes    t  ON o.Telescope_ID   = t.Telescope_ID
  JOIN Observatories ob ON t.Observatory_ID = ob.Observatory_ID
GROUP BY ob.Country ORDER BY Total_Observations DESC;
```

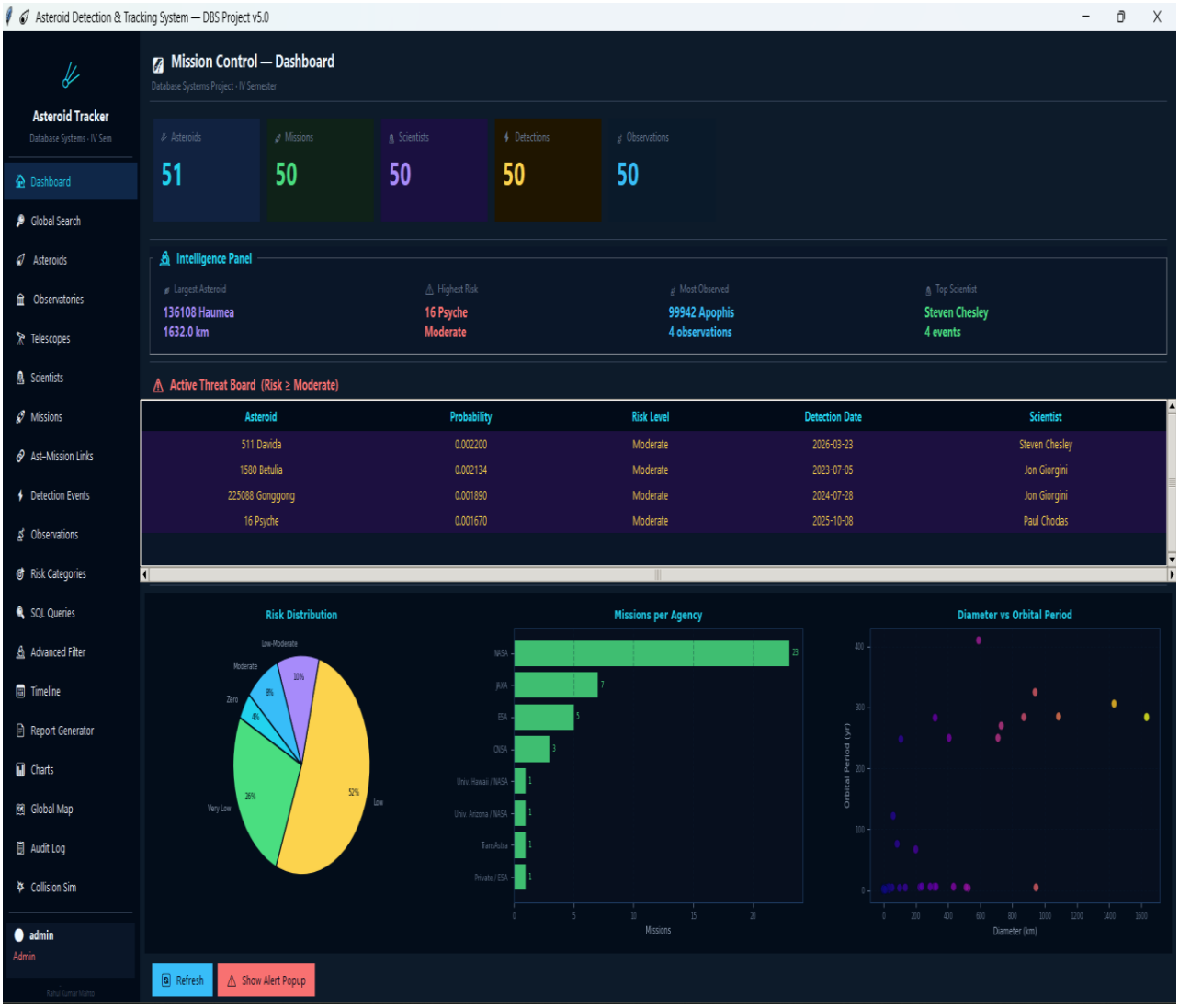
5.1 Query Capability Summary

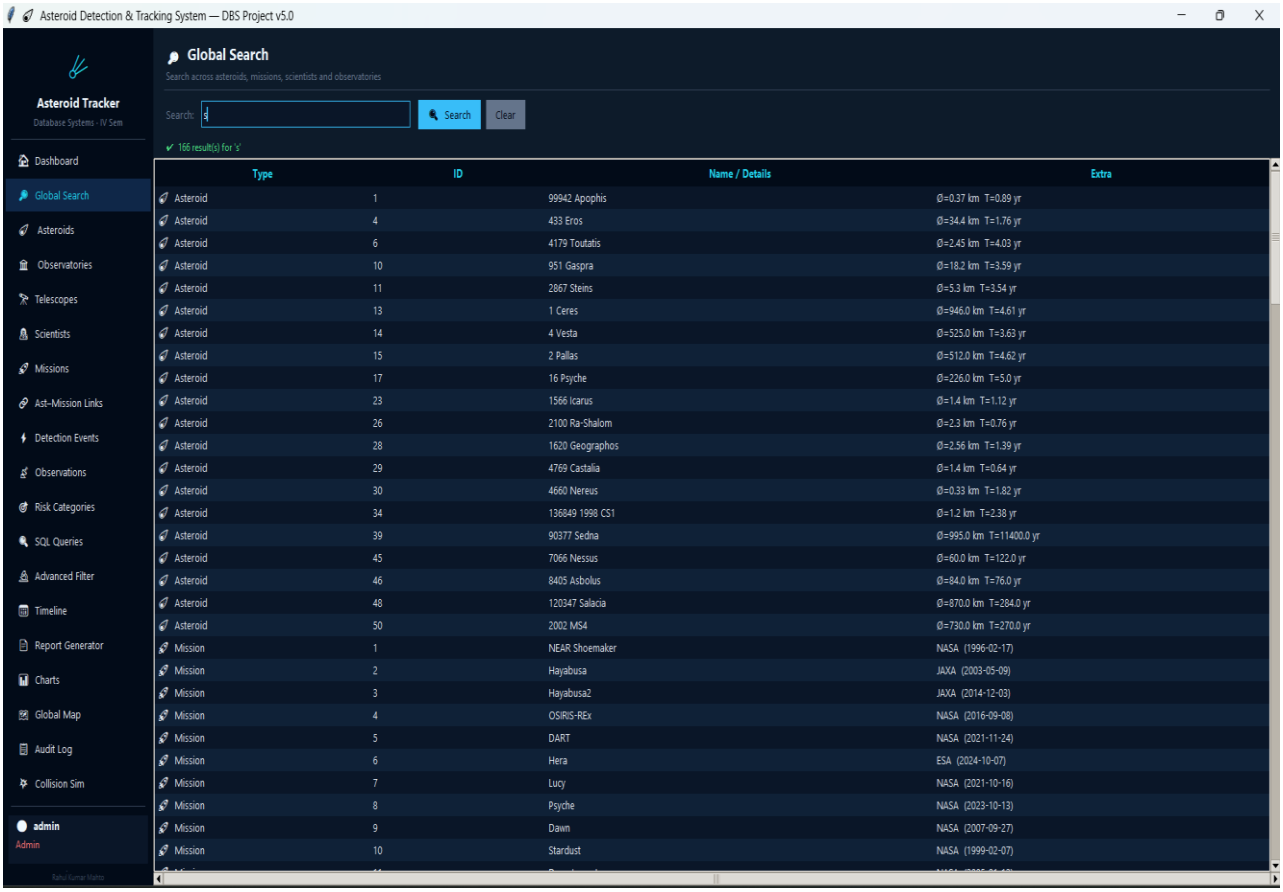
Q	Title	SQL Technique	Operational Value
Q1	High-Risk Events	Multi-JOIN, WHERE ≥ 5	Primary threat-board for PDCO
Q2	Telescope Productivity	LEFT JOIN, GROUP BY, COUNT	Resource allocation intelligence
Q3	Above-Average Risk	Scalar subquery, AVG	Statistical threat filtering
Q4	Avg Prob / Category	AVG, GROUP BY	Risk taxonomy validation
Q5	Asteroid–Mission Links	Junction table JOIN	Mission manifest reporting
Q6	Scientist Coverage	COUNT DISTINCT	Personnel productivity metrics
Q7	Undetected Asteroids	LEFT JOIN anti-join	Surveillance gap identification
Q8	Senior Scientist / Risk	Correlated subquery, MAX	QA for detection review
Q9	High-Risk Summary View	5-table JOIN, WHERE	Executive threat briefing
Q10	Observations by Country	3-hop join chain	Global network coverage map

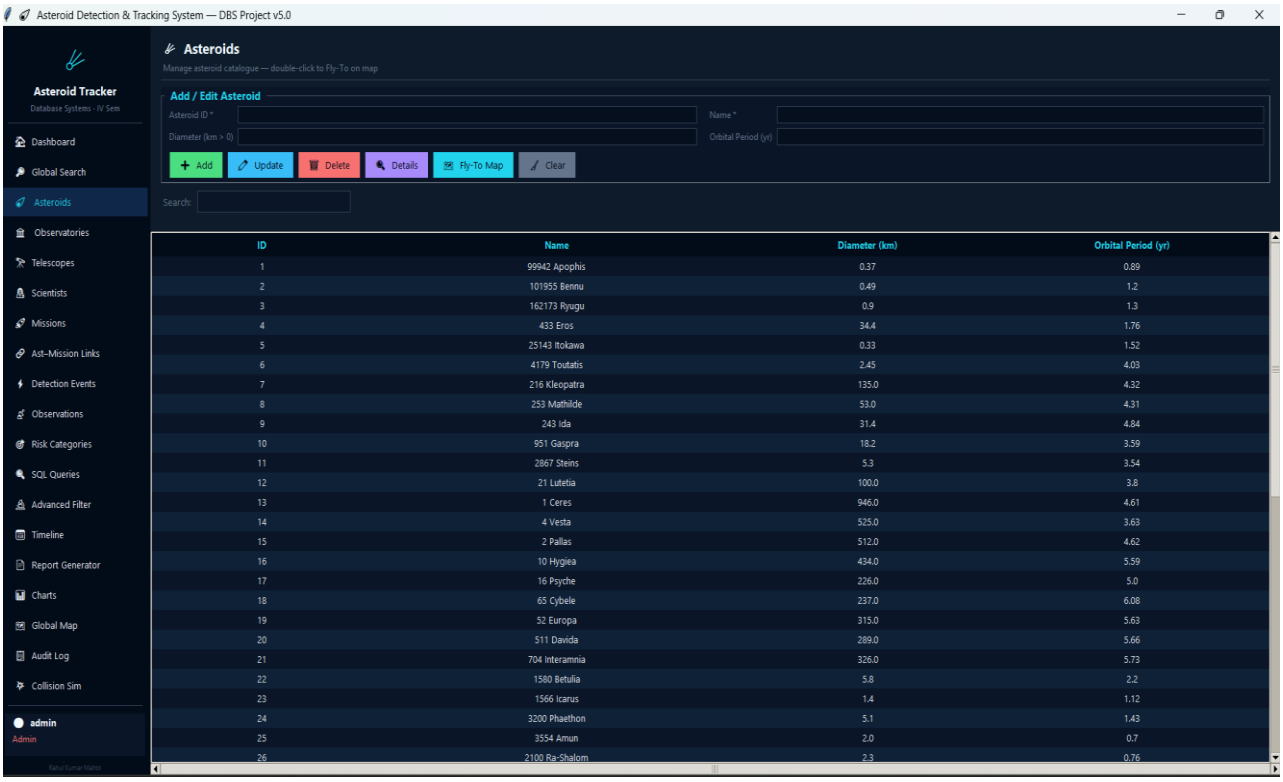
SNAPSHOTS

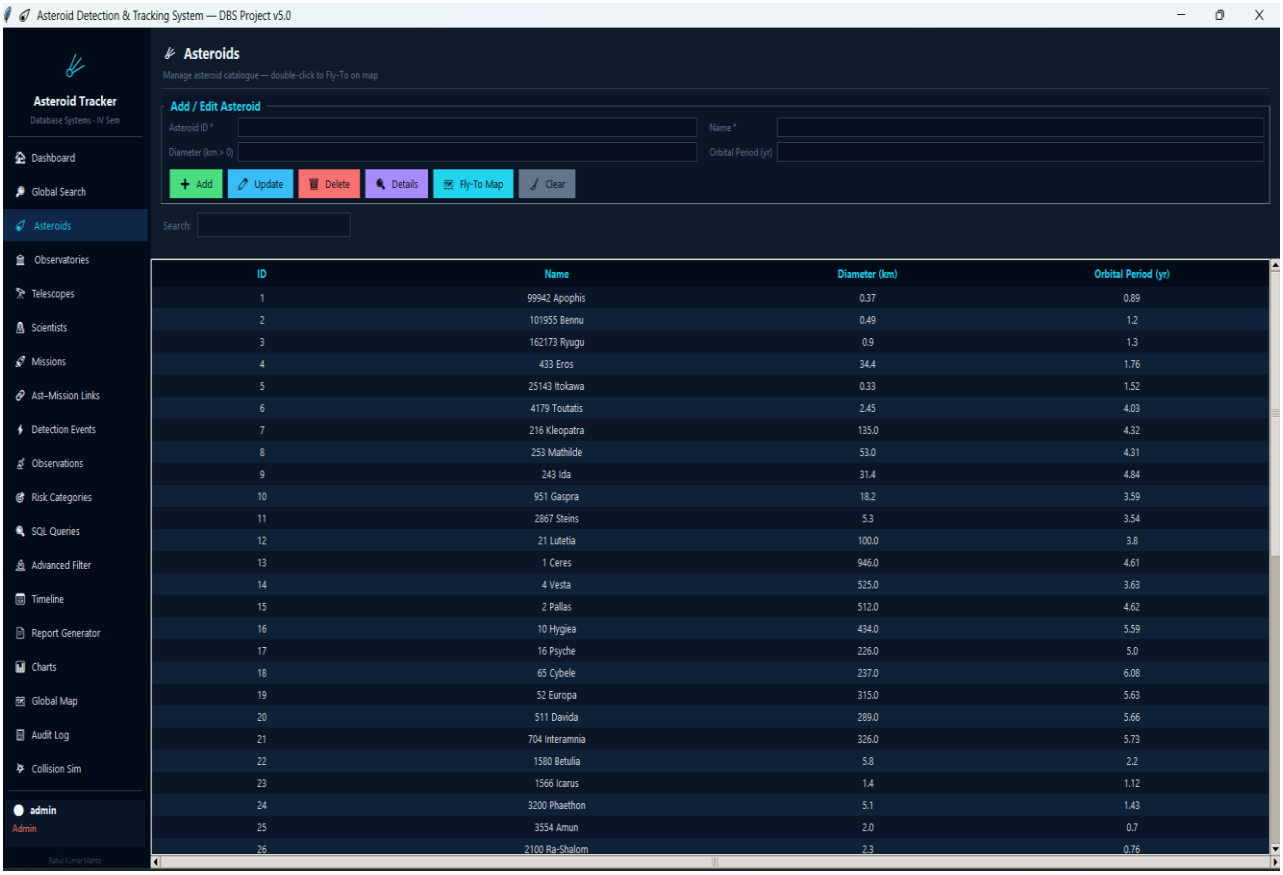

Planetary Defence Alert
RISK DETECTION ALERT
The following asteroids have elevated impact probability:
511 Davida Moderate P=0.002200
1580 Betulia Moderate P=0.002134
225088 Gonggong Moderate P=0.001890
16 Psyche Moderate P=0.001670
Acknowledge

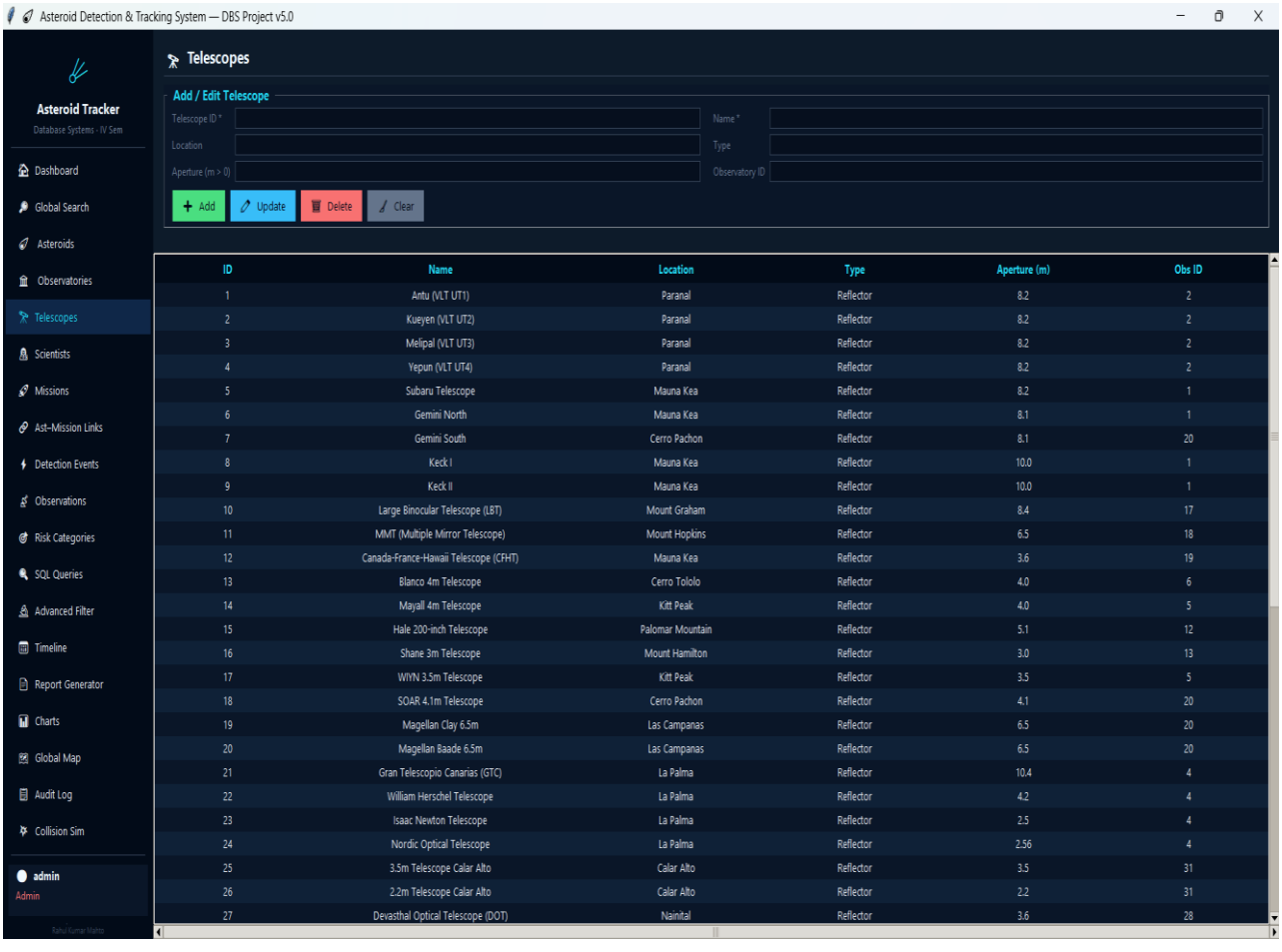

Asteroid Tracker — Login
Asteroid Detection & Tracking
Database Systems Project · IV Semester
Username
Password
Login

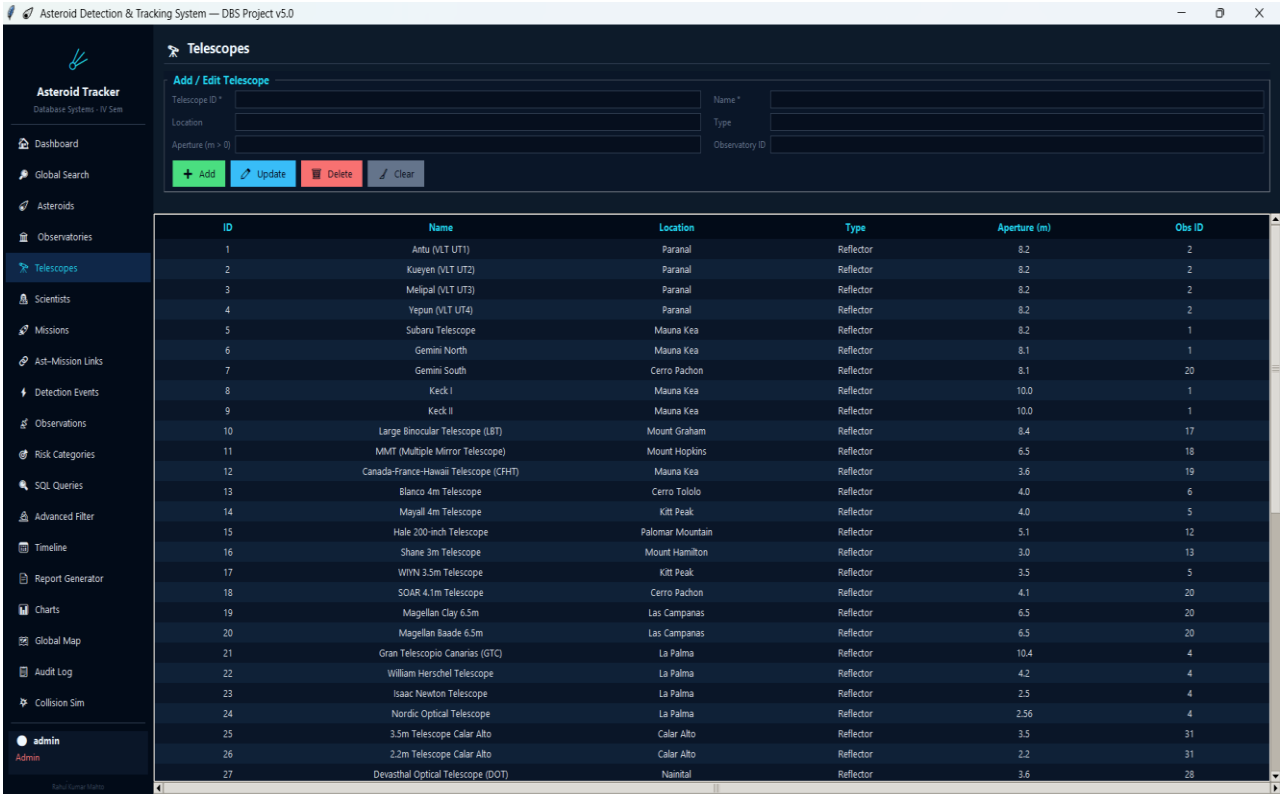

Asteroid Detection & Tracking System — DBS Project v5.0
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Global Search
Asteroids
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Scientists
Missions
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Risk Categories
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Advanced Filter
Timeline
Report Generator
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Audit Log
Collision Sim
admin
Admin
Mission Control — Dashboard
Database Systems Project · IV Semester
Asteroids 51
Missions 50
Scientists 50
Detections 50
Observations 50
Intelligence Panel
Largest Asteroid 136108 Haumea 1632.0 km
Highest Risk 16 Psyche Moderate
Most Observed 99942 Apophis 4 observations
Top Scientist Steven Chesley 4 events
Active Threat Board (Risk ≥ Moderate)
Asteroid Probability Risk Level Detection Date Scientist
511 Davida 0.002200 Moderate 2026-03-23 Steven Chesley
1580 Betulia 0.002134 Moderate 2023-07-05 Jon Giorgini
225088 Gonggong 0.001890 Moderate 2024-07-28 Jon Giorgini
16 Psyche 0.001670 Moderate 2025-10-08 Paul Chodas
Risk Distribution
Missions per Agency
Diameter vs Orbital Period
Orbital Period (yr)
Diameter (km)
Refresh
Show Alert Popup

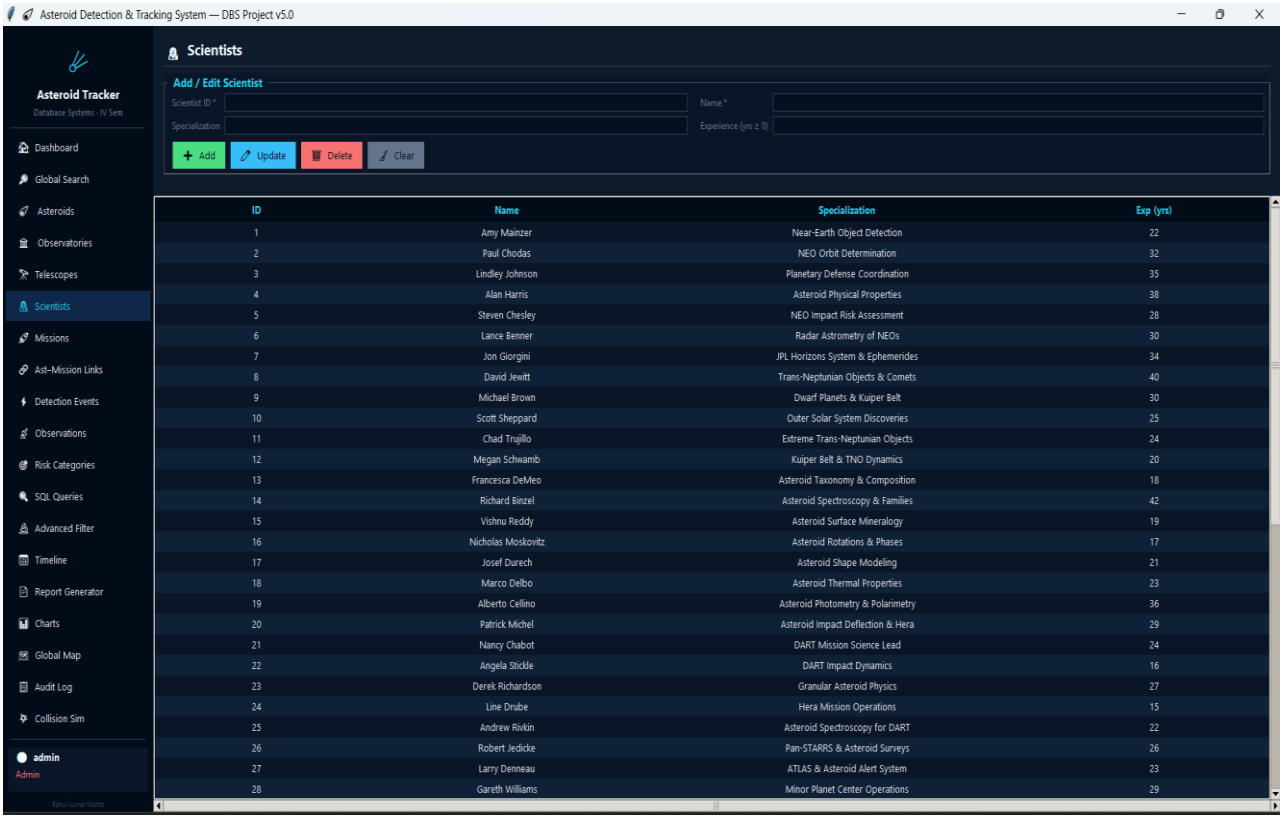

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Advanced Filter
Timeline
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Global Map
Audit Log
Collision Sim
admin
Admin
Global Search
Search across asteroids, missions, scientists and observatories
Search:
s
Search
Clear
166 result(s) for 's'
Type
ID
Name / Details
Extra
Asteroid 1 99942 Apophis Ø=0.37 km T=0.89 yr
Asteroid 4 433 Eros Ø=34.4 km T=1.76 yr
Asteroid 6 4179 Toutatis Ø=2.45 km T=4.03 yr
Asteroid 10 951 Gaspra Ø=18.2 km T=3.59 yr
Asteroid 11 2867 Steins Ø=5.3 km T=3.54 yr
Asteroid 13 1 Ceres Ø=946.0 km T=4.61 yr
Asteroid 14 4 Vesta Ø=525.0 km T=3.63 yr
Asteroid 15 2 Pallas Ø=512.0 km T=4.62 yr
Asteroid 17 16 Psyche Ø=226.0 km T=5.0 yr
Asteroid 23 1566 Icarus Ø=1.4 km T=1.12 yr
Asteroid 26 2100 Ra-Shalom Ø=2.3 km T=0.76 yr
Asteroid 28 1620 Geographos Ø=2.56 km T=1.39 yr
Asteroid 29 4769 Castalia Ø=1.4 km T=0.64 yr
Asteroid 30 4660 Nereus Ø=0.33 km T=1.82 yr
Asteroid 34 136849 1998 CS1 Ø=1.2 km T=2.38 yr
Asteroid 39 90377 Sedna Ø=995.0 km T=11400.0 yr
Asteroid 45 7066 Nessus Ø=60.0 km T=122.0 yr
Asteroid 46 8405 Asbolus Ø=84.0 km T=76.0 yr
Asteroid 48 120347 Salacia Ø=870.0 km T=284.0 yr
Asteroid 50 2002 MS4 Ø=730.0 km T=270.0 yr
Mission 1 NEAR Shoemaker NASA (1996-02-17)
Mission 2 Hayabusa JAXA (2003-05-09)
Mission 3 Hayabusa2 JAXA (2014-12-03)
Mission 4 OSIRIS-REx NASA (2016-09-08)
Mission 5 DART NASA (2021-11-24)
Mission 6 Hera ESA (2024-10-07)
Mission 7 Lucy NASA (2021-10-16)
Mission 8 Psyche NASA (2023-10-13)
Mission 9 Dawn NASA (2007-09-27)
Mission 10 Stardust NASA (1999-02-07)

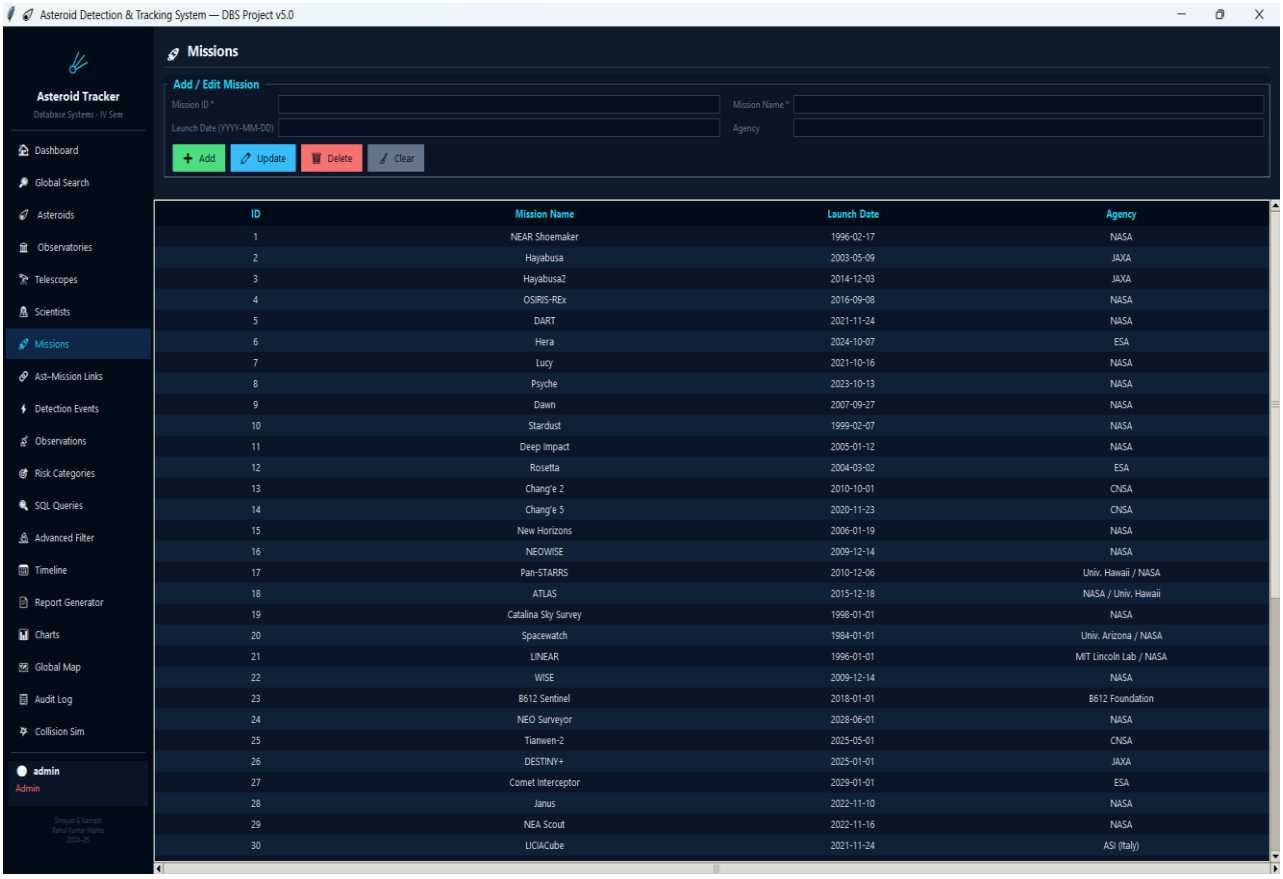

Asteroid Detection & Tracking System — DBS Project v5.0
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Audit Log
Collision Sim
admin
Admin
Asteroids
Manage asteroid catalogue — double-click to Fly-To on map
Add / Edit Asteroid
Asteroid ID *
Name *
Diameter (km > 0)
Orbital Period (yr)
Add
Update
Delete
Details
Fly-To Map
Clear
Search:
ID
Name
Diameter (km)
Orbital Period (yr)
1 99942 Apophis 0.37 0.89
2 101955 Bennu 0.49 1.2
3 162173 Ryugu 0.9 1.3
4 433 Eros 34.4 1.76
5 25143 Itokawa 0.33 1.52
6 4179 Toutatis 2.45 4.03
7 216 Kleopatra 135.0 4.32
8 253 Mathilde 53.0 4.31
9 243 Ida 31.4 4.84
10 951 Gaspra 18.2 3.59
11 2867 Steins 5.3 3.54
12 21 Lutetia 100.0 3.8
13 1 Ceres 946.0 4.61
14 4 Vesta 525.0 3.63
15 2 Pallas 512.0 4.62
16 10 Hygiea 434.0 5.59
17 16 Psyche 226.0 5.0
18 65 Cybele 237.0 6.08
19 52 Europa 315.0 5.63
20 511 Davida 289.0 5.66
21 704 Interamnia 326.0 5.73
22 1580 Betulia 5.8 2.2
23 1566 Icarus 1.4 1.12
24 3200 Phaethon 5.1 1.43
25 3554 Amun 2.0 0.7
26 2100 Ra-Shalom 2.3 0.76

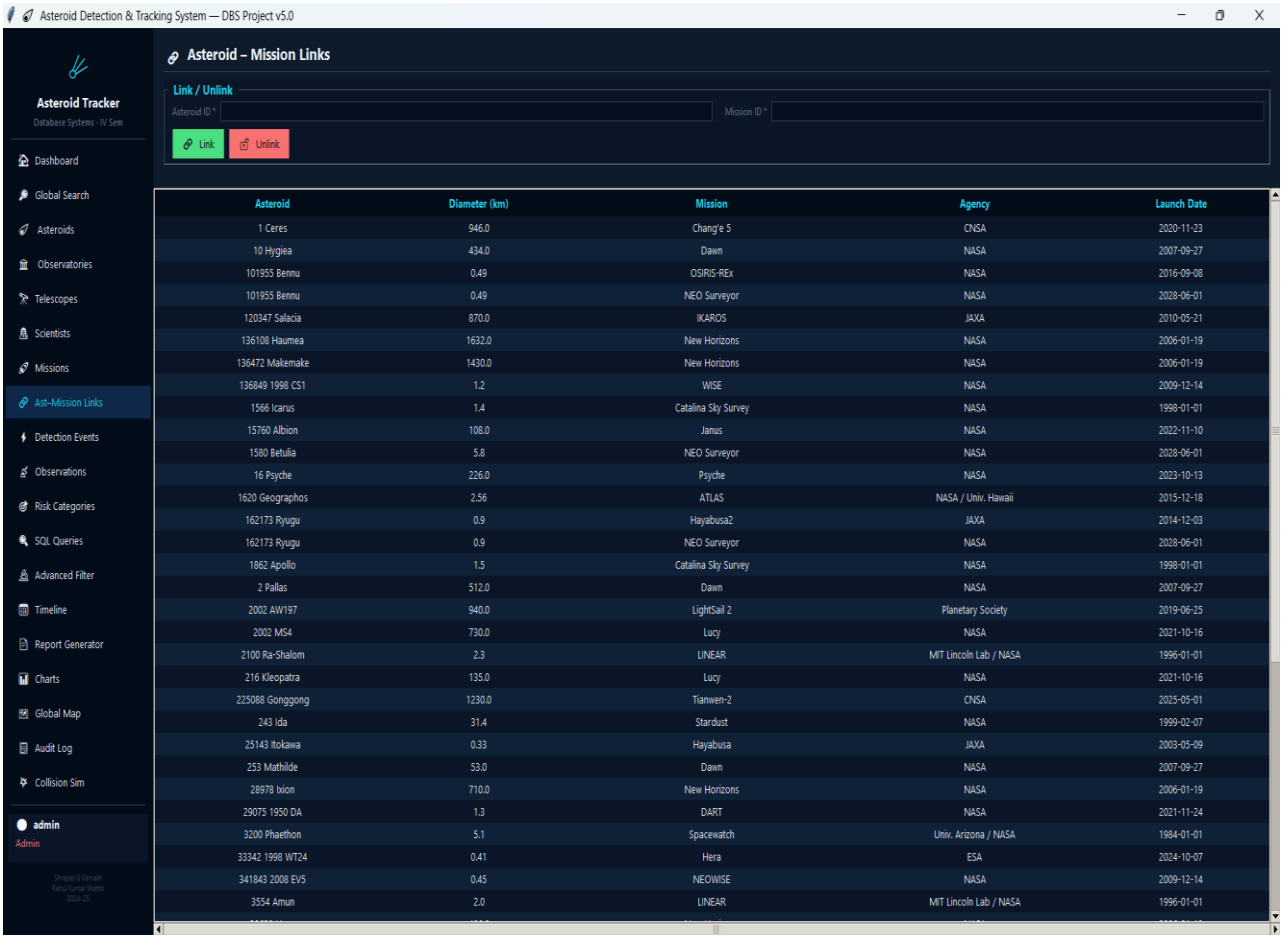

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admin
Admin
Asteroids
Manage asteroid catalogue — double-click to Fly-To on map
Add / Edit Asteroid
Asteroid ID *
Name *
Diameter (km > 0)
Orbital Period (yr)
Add
Update
Delete
Details
Fly-To Map
Clear
Search:
ID
Name
Diameter (km)
Orbital Period (yr)
1 99942 Apophis 0.37 0.89
2 101955 Bennu 0.49 1.2
3 162173 Ryugu 0.9 1.3
4 433 Eros 34.4 1.76
5 25143 Itokawa 0.33 1.52
6 4179 Toutatis 2.45 4.03
7 216 Kleopatra 135.0 4.32
8 253 Mathilde 53.0 4.31
9 243 Ida 31.4 4.84
10 951 Gaspra 18.2 3.59
11 2867 Steins 5.3 3.54
12 21 Lutetia 100.0 3.8
13 1 Ceres 946.0 4.61
14 4 Vesta 525.0 3.63
15 2 Pallas 512.0 4.62
16 10 Hygiea 434.0 5.59
17 16 Psyche 226.0 5.0
18 65 Cybele 237.0 6.08
19 52 Europa 315.0 5.63
20 511 Davida 289.0 5.66
21 704 Interamnia 326.0 5.73
22 1580 Betulia 5.8 2.2
23 1566 Icarus 1.4 1.12
24 3200 Phaethon 5.1 1.43
25 3554 Amun 2.0 0.7
26 2100 Ra-Shalom 2.3 0.76

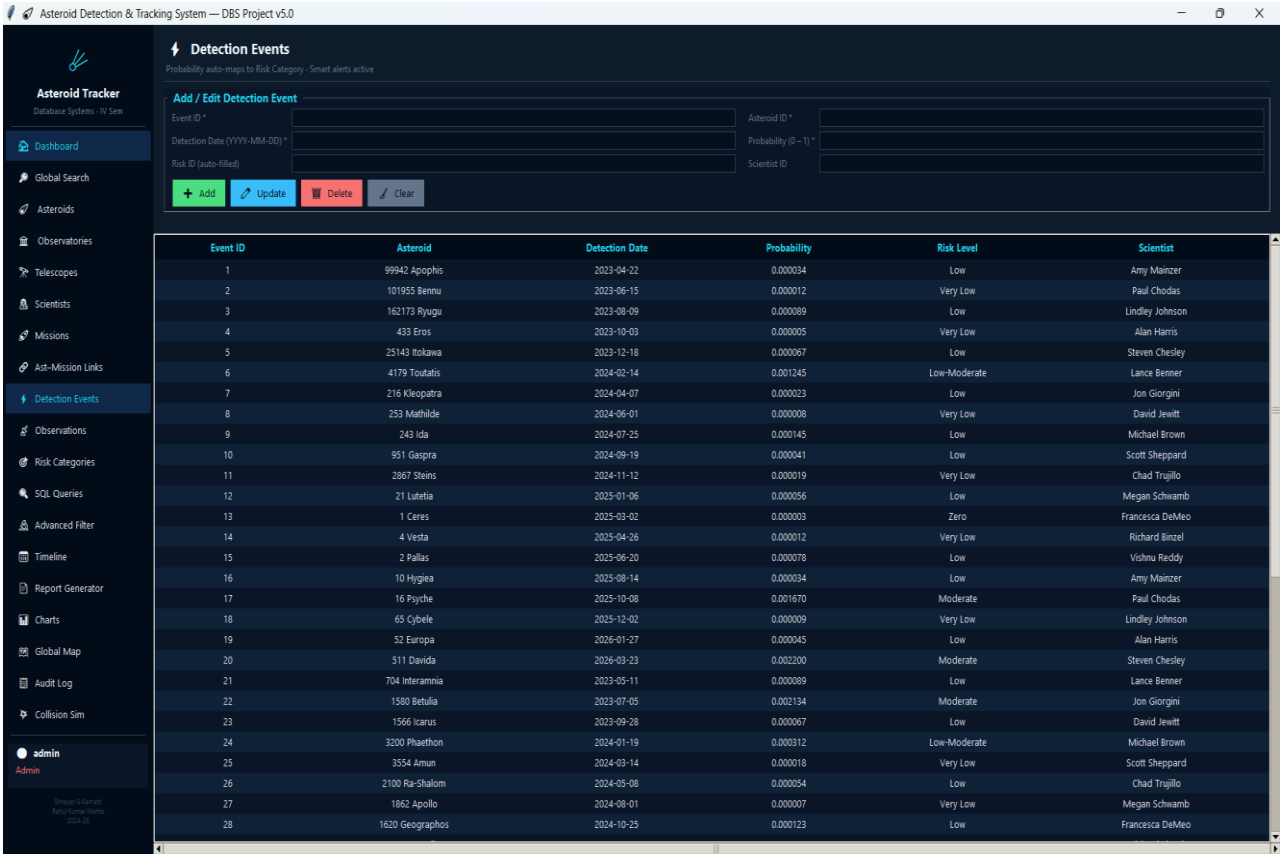

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admin
Admin
Telescopes
Add / Edit Telescope
Telescope ID *
Name *
Location
Type
Aperture (m > 0)
Observatory ID
Add
Update
Delete
Clear
ID
Name
Location
Type
Aperture (m)
Obs ID
1 Antu (VLT UT1) Paranal Reflector 8.2 2
2 Kueyen (VLT UT2) Paranal Reflector 8.2 2
3 Melipal (VLT UT3) Paranal Reflector 8.2 2
4 Yepun (VLT UT4) Paranal Reflector 8.2 2
5 Subaru Telescope Mauna Kea Reflector 8.2 1
6 Gemini North Mauna Kea Reflector 8.1 1
7 Gemini South Cerro Pachon Reflector 8.1 20
8 Keck I Mauna Kea Reflector 10.0 1
9 Keck II Mauna Kea Reflector 10.0 1
10 Large Binocular Telescope (LBT) Mount Graham Reflector 8.4 17
11 MMT (Multiple Mirror Telescope) Mount Hopkins Reflector 6.5 18
12 Canada-France-Hawaii Telescope (CFHT) Mauna Kea Reflector 3.6 19
13 Blanco 4m Telescope Cerro Tololo Reflector 4.0 6
14 Mayall 4m Telescope Kitt Peak Reflector 4.0 5
15 Hale 200-inch Telescope Palomar Mountain Reflector 5.1 12
16 Shane 3m Telescope Mount Hamilton Reflector 3.0 13
17 WIYN 3.5m Telescope Kitt Peak Reflector 3.5 5
18 SOAR 4.1m Telescope Cerro Pachon Reflector 4.1 20
19 Magellan Clay 6.5m Las Campanas Reflector 6.5 20
20 Magellan Baade 6.5m Las Campanas Reflector 6.5 20
21 Gran Telescopio Canarias (GTC) La Palma Reflector 10.4 4
22 William Herschel Telescope La Palma Reflector 4.2 4
23 Isaac Newton Telescope La Palma Reflector 2.5 4
24 Nordic Optical Telescope La Palma Reflector 2.56 4
25 3.5m Telescope Calar Alto Calar Alto Reflector 3.5 31
26 2.2m Telescope Calar Alto Calar Alto Reflector 2.2 31
27 Devasthal Optical Telescope (DOT) Nainital Reflector 3.6 28

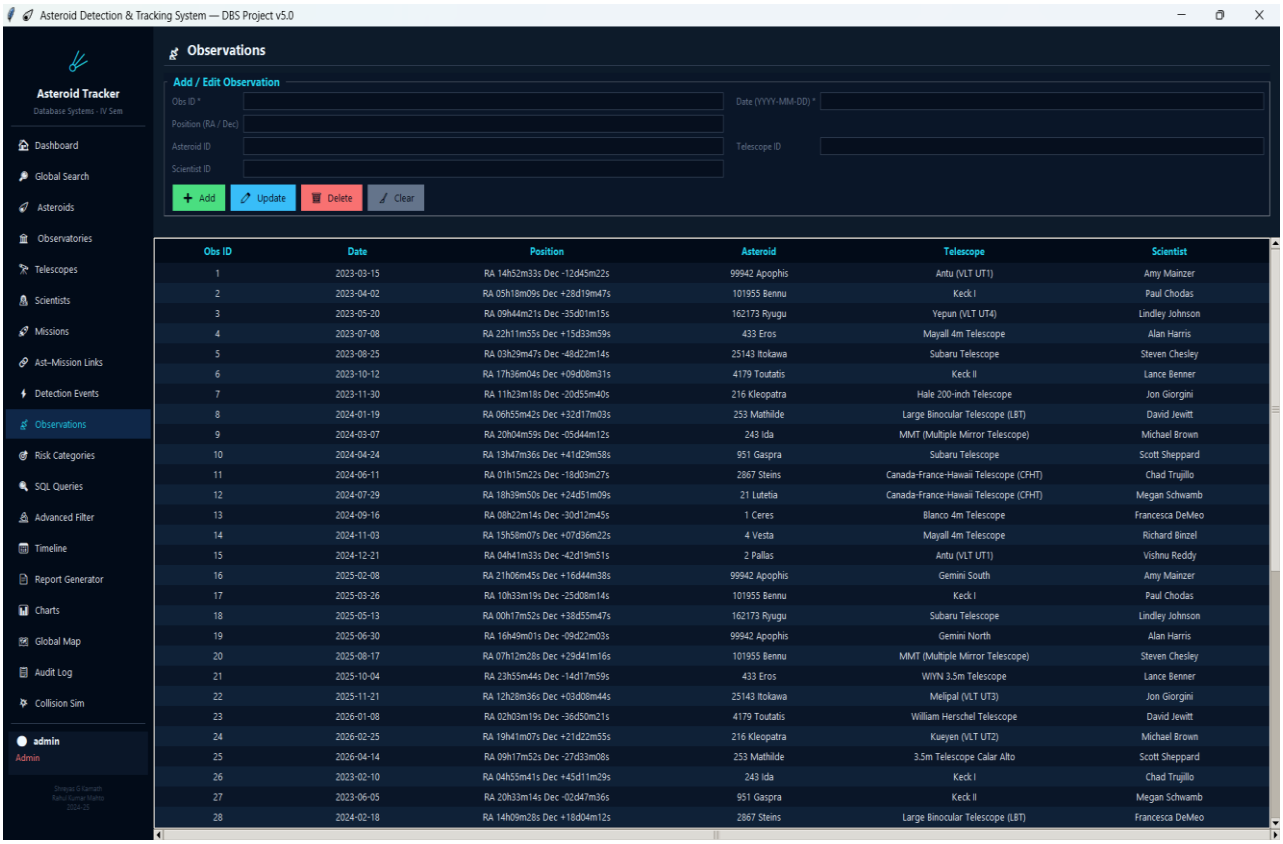

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Admin
Telescopes
Add / Edit Telescope
Telescope ID *
Name *
Location
Type
Aperture (m > 0)
Observatory ID
Add
Update
Delete
Clear
ID Name Location Type Aperture (m) Obs ID
1 Antu (VLT UT1) Paranal Reflector 8.2 2
2 Kueyen (VLT UT2) Paranal Reflector 8.2 2
3 Melipal (VLT UT3) Paranal Reflector 8.2 2
4 Yepun (VLT UT4) Paranal Reflector 8.2 2
5 Subaru Telescope Mauna Kea Reflector 8.2 1
6 Gemini North Mauna Kea Reflector 8.1 1
7 Gemini South Cerro Pachon Reflector 8.1 20
8 Keck I Mauna Kea Reflector 10.0 1
9 Keck II Mauna Kea Reflector 10.0 1
10 Large Binocular Telescope (LBT) Mount Graham Reflector 8.4 17
11 MMT (Multiple Mirror Telescope) Mount Hopkins Reflector 6.5 18
12 Canada-France-Hawaii Telescope (CFHT) Mauna Kea Reflector 3.6 19
13 Blanco 4m Telescope Cerro Tololo Reflector 4.0 6
14 Mayall 4m Telescope Kitt Peak Reflector 4.0 5
15 Hale 200-inch Telescope Palomar Mountain Reflector 5.1 12
16 Shane 3m Telescope Mount Hamilton Reflector 3.0 13
17 WIYN 3.5m Telescope Kitt Peak Reflector 3.5 5
18 SOAR 4.1m Telescope Cerro Pachon Reflector 4.1 20
19 Magellan Clay 6.5m Las Campanas Reflector 6.5 20
20 Magellan Baade 6.5m Las Campanas Reflector 6.5 20
21 Gran Telescopio Canarias (GTC) La Palma Reflector 10.4 4
22 William Herschel Telescope La Palma Reflector 4.2 4
23 Isaac Newton Telescope La Palma Reflector 2.5 4
24 Nordic Optical Telescope La Palma Reflector 2.56 4
25 3.5m Telescope Calar Alto Calar Alto Reflector 3.5 31
26 2.2m Telescope Calar Alto Calar Alto Reflector 2.2 31
27 Devasthal Optical Telescope (DOT) Nainital Reflector 3.6 28

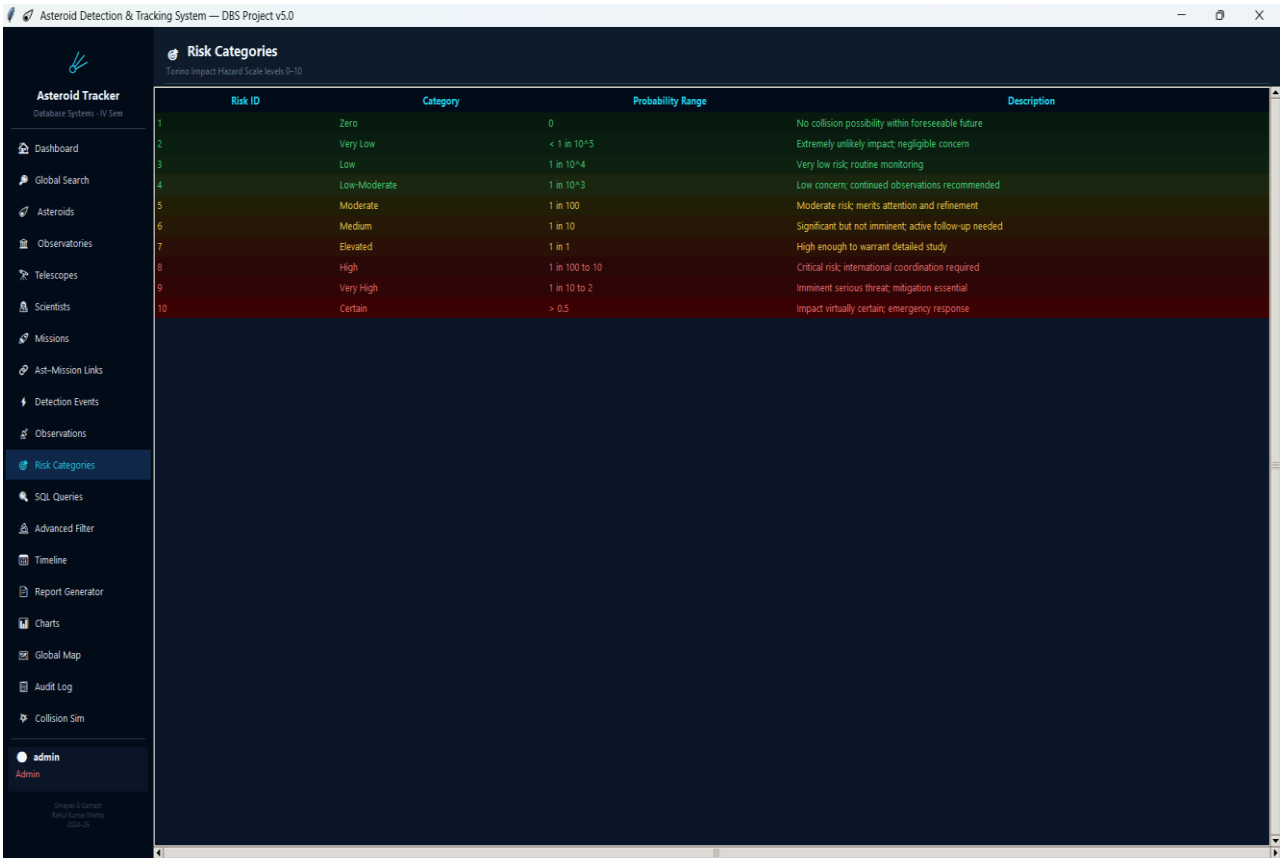

Asteroid Detection & Tracking System — DBS Project v5.0
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Audit Log
Collision Sim
admin
Admin
Scientists
Add / Edit Scientist
Scientist ID *
Name *
Specialization
Experience (yrs ≥ 0)
Add
Update
Delete
Clear
ID Name Specialization Exp (yrs)
1 Amy Mainzer Near-Earth Object Detection 22
2 Paul Chodas NEO Orbit Determination 32
3 Lindley Johnson Planetary Defense Coordination 35
4 Alan Harris Asteroid Physical Properties 38
5 Steven Chesley NEO Impact Risk Assessment 28
6 Lance Benner Radar Astrometry of NEOs 30
7 Jon Giorgini JPL Horizons System & Ephemerides 34
8 David Jewitt Trans-Neptunian Objects & Comets 40
9 Michael Brown Dwarf Planets & Kuiper Belt 30
10 Scott Sheppard Outer Solar System Discoveries 25
11 Chad Trujillo Extreme Trans-Neptunian Objects 24
12 Megan Schwamb Kuiper Belt & TNO Dynamics 20
13 Francesca DeMeo Asteroid Taxonomy & Composition 18
14 Richard Binzel Asteroid Spectroscopy & Families 42
15 Vishnu Reddy Asteroid Surface Mineralogy 19
16 Nicholas Moskovitz Asteroid Rotations & Phases 17
17 Josef Durech Asteroid Shape Modeling 21
18 Marco Delbo Asteroid Thermal Properties 23
19 Alberto Cellino Asteroid Photometry & Polarimetry 36
20 Patrick Michel Asteroid Impact Deflection & Hera 29
21 Nancy Chabot DART Mission Science Lead 24
22 Angela Stickle DART Impact Dynamics 16
23 Derek Richardson Granular Asteroid Physics 27
24 Line Drube Hera Mission Operations 15
25 Andrew Rivkin Asteroid Spectroscopy for DART 22
26 Robert Jedicke Pan-STARRS & Asteroid Surveys 26
27 Larry Denneau ATLAS & Asteroid Alert System 23
28 Gareth Williams Minor Planet Center Operations 29

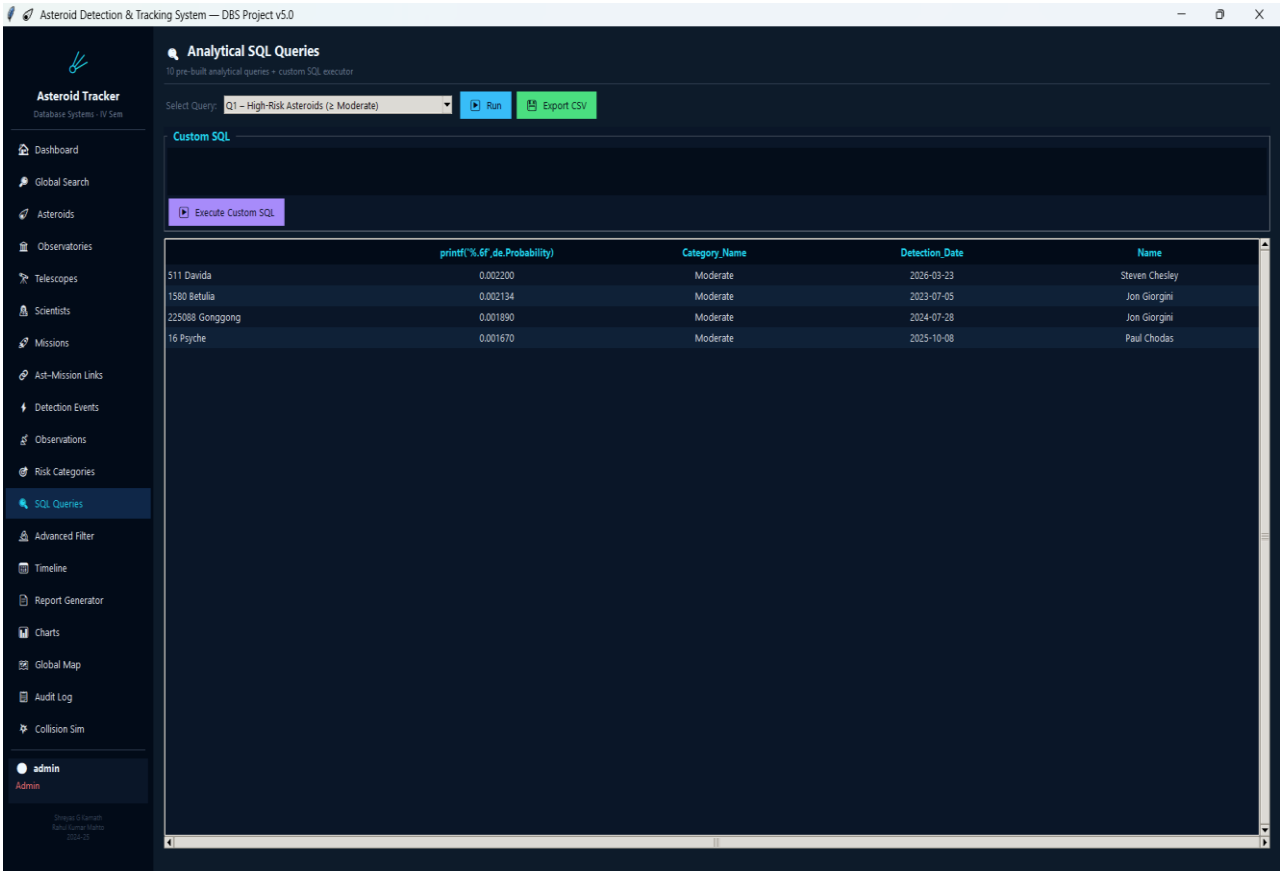

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admin
Admin
Missions
Add / Edit Mission
Mission ID *
Mission Name *
Launch Date (YYYY-MM-DD)
Agency
Add
Update
Delete
Clear
ID
Mission Name
Launch Date
Agency
1 NEAR Shoemaker 1996-02-17 NASA
2 Hayabusa 2003-05-09 JAXA
3 Hayabusa2 2014-12-03 JAXA
4 OSIRIS-REx 2016-09-08 NASA
5 DART 2021-11-24 NASA
6 Hera 2024-10-07 ESA
7 Lucy 2021-10-16 NASA
8 Psyche 2023-10-13 NASA
9 Dawn 2007-09-27 NASA
10 Stardust 1999-02-07 NASA
11 Deep Impact 2005-01-12 NASA
12 Rosetta 2004-03-02 ESA
13 Chang'e 2 2010-10-01 CNSA
14 Chang'e 5 2020-11-23 CNSA
15 New Horizons 2006-01-19 NASA
16 NEOWISE 2009-12-14 NASA
17 Pan-STARRS 2010-12-06 Univ. Hawaii / NASA
18 ATLAS 2015-12-18 NASA / Univ. Hawaii
19 Catalina Sky Survey 1998-01-01 NASA
20 Spacewatch 1984-01-01 Univ. Arizona / NASA
21 LINEAR 1996-01-01 MIT Lincoln Lab / NASA
22 WISE 2009-12-14 NASA
23 B612 Sentinel 2018-01-01 B612 Foundation
24 NEO Surveyor 2028-06-01 NASA
25 Tianwen-2 2025-05-01 CNSA
26 DESTINY+ 2025-01-01 JAXA
27 Comet Interceptor 2029-01-01 ESA
28 Janus 2022-11-10 NASA
29 NEA Scout 2022-11-16 NASA
30 LICIACube 2021-11-24 ASI (Italy)

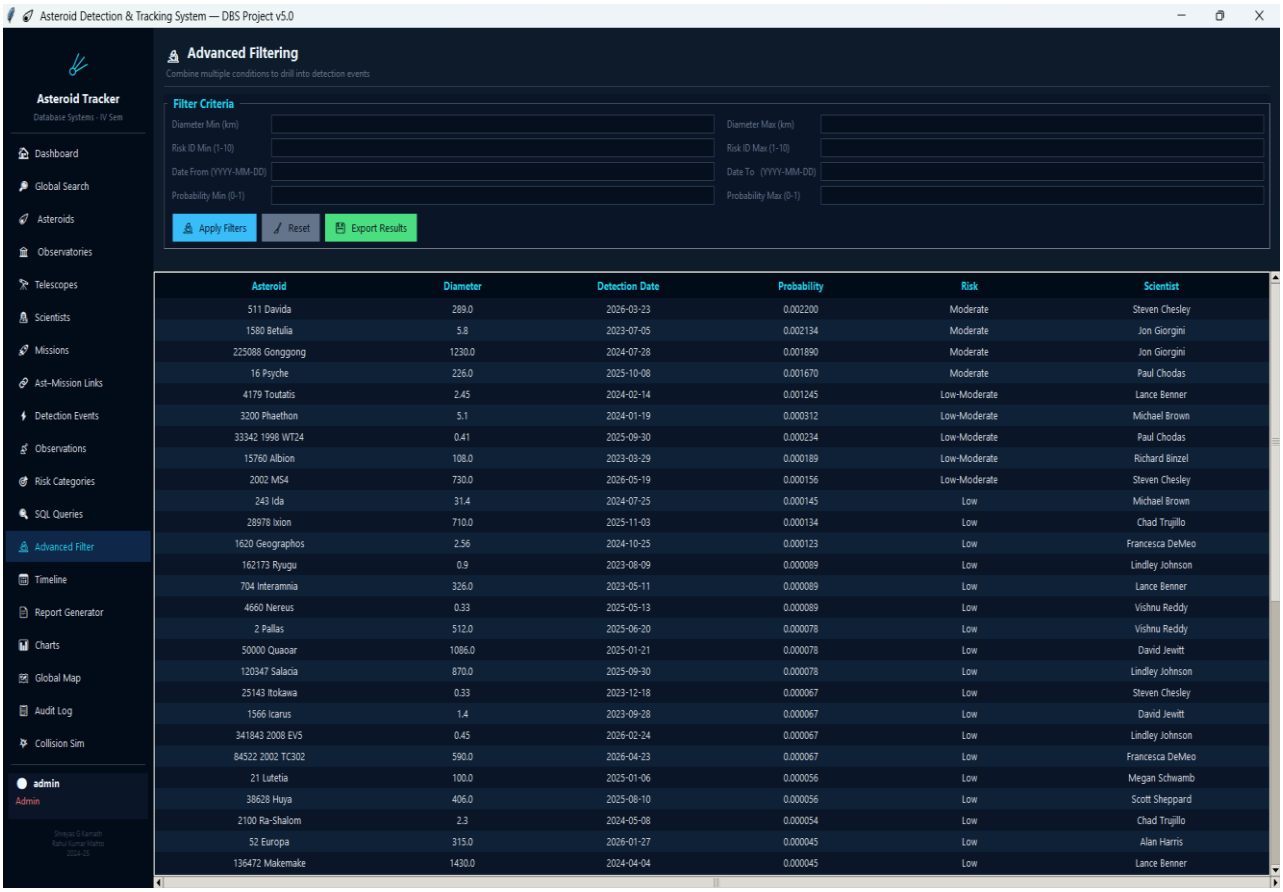

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Admin
Asteroid – Mission Links
Link / Unlink
Asteroid ID *
Mission ID *
Link
Unlink
Asteroid
Diameter (km)
Mission
Agency
Launch Date
1 Ceres 946.0 Chang'e 5 CNSA 2020-11-23
10 Hygiea 434.0 Dawn NASA 2007-09-27
101955 Bennu 0.49 OSIRIS-REx NASA 2016-09-08
101955 Bennu 0.49 NEO Surveyor NASA 2028-06-01
120347 Salacia 870.0 IKAROS JAXA 2010-05-21
136108 Haumea 1632.0 New Horizons NASA 2006-01-19
136472 Makemake 1430.0 New Horizons NASA 2006-01-19
136849 1998 CS1 1.2 WISE NASA 2009-12-14
1566 Icarus 1.4 Catalina Sky Survey NASA 1998-01-01
15760 Albion 108.0 Janus NASA 2022-11-10
1580 Betulia 5.8 NEO Surveyor NASA 2028-06-01
16 Psyche 226.0 Psyche NASA 2023-10-13
1620 Geographos 2.56 ATLAS NASA / Univ. Hawaii 2015-12-18
162173 Ryugu 0.9 Hayabusa2 JAXA 2014-12-03
162173 Ryugu 0.9 NEO Surveyor NASA 2028-06-01
1862 Apollo 1.5 Catalina Sky Survey NASA 1998-01-01
2 Pallas 512.0 Dawn NASA 2007-09-27
2002 AW197 940.0 LightSail 2 Planetary Society 2019-06-25
2002 MS4 730.0 Lucy NASA 2021-10-16
2100 Ra-Shalom 2.3 LINEAR MIT Lincoln Lab / NASA 1996-01-01
216 Kleopatra 135.0 Lucy NASA 2021-10-16
225088 Gonggong 1230.0 Tianwen-2 CNSA 2025-05-01
243 Ida 31.4 Stardust NASA 1999-02-07
25143 Itokawa 0.33 Hayabusa JAXA 2003-05-09
253 Mathilde 53.0 Dawn NASA 2007-09-27
28978 Ixion 710.0 New Horizons NASA 2006-01-19
29075 1950 DA 1.3 DART NASA 2021-11-24
3200 Phaethon 5.1 Spacewatch Univ. Arizona / NASA 1984-01-01
33342 1998 WT24 0.41 Hera ESA 2024-10-07
341843 2008 EV5 0.45 NEOWISE NASA 2009-12-14
3554 Amun 2.0 LINEAR MIT Lincoln Lab / NASA 1996-01-01

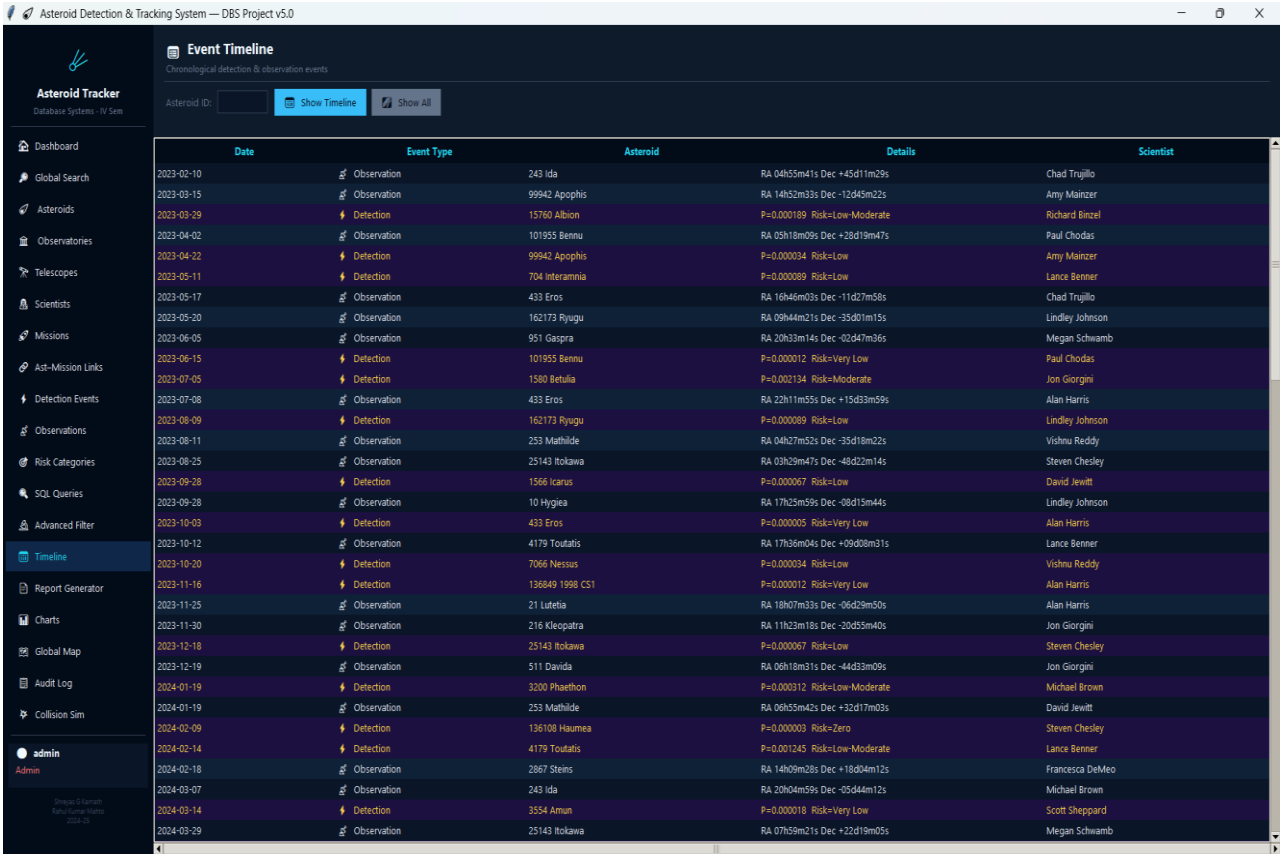

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admin
Admin
Detection Events
Probability auto-maps to Risk Category · Smart alerts active
Add / Edit Detection Event
Event ID *
Asteroid ID *
Detection Date (YYYY-MM-DD) *
Probability (0 – 1) *
Risk ID (auto-filled)
Scientist ID
Add
Update
Delete
Clear
Event ID
Asteroid
Detection Date
Probability
Risk Level
Scientist
1 99942 Apophis 2023-04-22 0.000034 Low Amy Mainzer
2 101955 Bennu 2023-06-15 0.000012 Very Low Paul Chodas
3 162173 Ryugu 2023-08-09 0.000089 Low Lindley Johnson
4 433 Eros 2023-10-03 0.000005 Very Low Alan Harris
5 25143 Itokawa 2023-12-18 0.000067 Low Steven Chesley
6 4179 Toutatis 2024-02-14 0.001245 Low-Moderate Lance Benner
7 216 Kleopatra 2024-04-07 0.000023 Low Jon Giorgini
8 253 Mathilde 2024-06-01 0.000008 Very Low David Jewitt
9 243 Ida 2024-07-25 0.000145 Low Michael Brown
10 951 Gaspra 2024-09-19 0.000041 Low Scott Sheppard
11 2867 Steins 2024-11-12 0.000019 Very Low Chad Trujillo
12 21 Lutetia 2025-01-06 0.000056 Low Megan Schwamb
13 1 Ceres 2025-03-02 0.000003 Zero Francesca DeMeo
14 4 Vesta 2025-04-26 0.000012 Very Low Richard Binzel
15 2 Pallas 2025-06-20 0.000078 Low Vishnu Reddy
16 10 Hygiea 2025-08-14 0.000034 Low Amy Mainzer
17 16 Psyche 2025-10-08 0.001670 Moderate Paul Chodas
18 65 Cybele 2025-12-02 0.000009 Very Low Lindley Johnson
19 52 Europa 2026-01-27 0.000045 Low Alan Harris
20 511 Davida 2026-03-23 0.002200 Moderate Steven Chesley
21 704 Interamnia 2023-05-11 0.000089 Low Lance Benner
22 1580 Betulia 2023-07-05 0.002134 Moderate Jon Giorgini
23 1566 Icarus 2023-09-28 0.000067 Low David Jewitt
24 3200 Phaethon 2024-01-19 0.000312 Low-Moderate Michael Brown
25 3554 Amun 2024-03-14 0.000018 Very Low Scott Sheppard
26 2100 Ra-Shalom 2024-05-08 0.000054 Low Chad Trujillo
27 1862 Apollo 2024-08-01 0.000007 Very Low Megan Schwamb
28 1620 Geographos 2024-10-25 0.000123 Low Francesca DeMeo

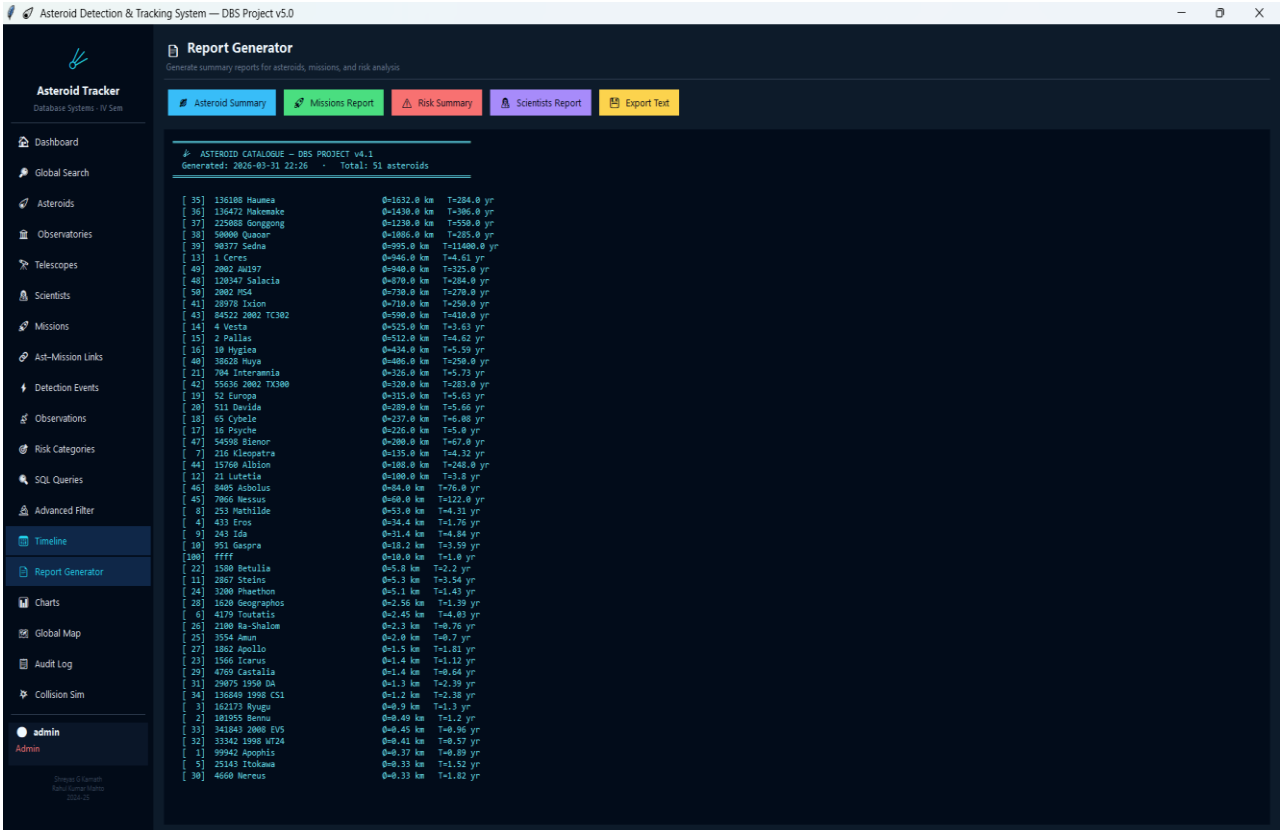

Asteroid Detection & Tracking System — DBS Project v5.0
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Audit Log
Collision Sim
admin
Admin
Observations
Add / Edit Observation
Obs ID *
Date (YYYY-MM-DD) *
Position (RA / Dec)
Asteroid ID
Telescope ID
Scientist ID
Add
Update
Delete
Clear
Obs ID
Date
Position
Asteroid
Telescope
Scientist
1 2023-03-15 RA 14h52m33s Dec -12d45m22s 99942 Apophis Antu (VLT UT1) Amy Mainzer
2 2023-04-02 RA 05h18m09s Dec +28d19m47s 101955 Bennu Keck I Paul Chodas
3 2023-05-20 RA 09h44m21s Dec -35d01m15s 162173 Ryugu Yepun (VLT UT4) Lindley Johnson
4 2023-07-08 RA 22h11m55s Dec +15d33m59s 433 Eros Mayall 4m Telescope Alan Harris
5 2023-08-25 RA 03h29m47s Dec -48d22m14s 25143 Itokawa Subaru Telescope Steven Chesley
6 2023-10-12 RA 17h36m04s Dec +09d08m31s 4179 Toutatis Keck II Lance Benner
7 2023-11-30 RA 11h23m18s Dec -20d55m40s 216 Kleopatra Hale 200-inch Telescope Jon Giorgini
8 2024-01-19 RA 06h55m42s Dec +32d17m03s 253 Mathilde Large Binocular Telescope (LBT) David Jewitt
9 2024-03-07 RA 20h04m59s Dec -05d44m12s 243 Ida MMT (Multiple Mirror Telescope) Michael Brown
10 2024-04-24 RA 13h47m36s Dec +41d29m58s 951 Gaspra Subaru Telescope Scott Sheppard
11 2024-06-11 RA 01h15m22s Dec -18d03m27s 2867 Steins Canada-France-Hawaii Telescope (CFHT) Chad Trujillo
12 2024-07-29 RA 18h39m50s Dec +24d51m09s 21 Lutetia Canada-France-Hawaii Telescope (CFHT) Megan Schwamb
13 2024-09-16 RA 08h22m14s Dec -30d12m45s 1 Ceres Blanco 4m Telescope Francesca DeMeo
14 2024-11-03 RA 15h58m07s Dec +07d36m22s 4 Vesta Mayall 4m Telescope Richard Binzel
15 2024-12-21 RA 04h41m33s Dec -42d19m51s 2 Pallas Antu (VLT UT1) Vishnu Reddy
16 2025-02-08 RA 21h06m45s Dec +16d44m38s 99942 Apophis Gemini South Amy Mainzer
17 2025-03-26 RA 10h33m19s Dec -25d08m14s 101955 Bennu Keck I Paul Chodas
18 2025-05-13 RA 00h17m52s Dec +38d55m47s 162173 Ryugu Subaru Telescope Lindley Johnson
19 2025-06-30 RA 16h49m01s Dec -09d22m03s 99942 Apophis Gemini North Alan Harris
20 2025-08-17 RA 07h12m28s Dec +29d41m16s 101955 Bennu MMT (Multiple Mirror Telescope) Steven Chesley
21 2025-10-04 RA 23h55m44s Dec -14d17m59s 433 Eros WIYN 3.5m Telescope Lance Benner
22 2025-11-21 RA 12h28m36s Dec +03d08m44s 25143 Itokawa Melipal (VLT UT3) Jon Giorgini
23 2026-01-08 RA 02h03m19s Dec -36d50m21s 4179 Toutatis William Herschel Telescope David Jewitt
24 2026-02-25 RA 19h41m07s Dec +21d22m55s 216 Kleopatra Kueyen (VLT UT2) Michael Brown
25 2026-04-14 RA 09h17m52s Dec -27d33m08s 253 Mathilde 3.5m Telescope Calar Alto Scott Sheppard
26 2023-02-10 RA 04h55m41s Dec +45d11m29s 243 Ida Keck I Chad Trujillo
27 2023-06-05 RA 20h33m14s Dec -02d47m36s 951 Gaspra Keck II Megan Schwamb
28 2024-02-18 RA 14h09m28s Dec +18d04m12s 2867 Steins Large Binocular Telescope (LBT) Francesca DeMeo


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Admin
Risk Categories
Torino Impact Hazard Scale levels 0–10
Risk ID
Category
Probability Range
Description
1 Zero 0 No collision possibility within foreseeable future
2 Very Low < 1 in 10^5 Extremely unlikely impact; negligible concern
3 Low 1 in 10^4 Very low risk; routine monitoring
4 Low-Moderate 1 in 10^3 Low concern; continued observations recommended
5 Moderate 1 in 100 Moderate risk; merits attention and refinement
6 Medium 1 in 10 Significant but not imminent; active follow-up needed
7 Elevated 1 in 1 High enough to warrant detailed study
8 High 1 in 100 to 10 Critical risk; international coordination required
9 Very High 1 in 10 to 2 Imminent serious threat; mitigation essential
10 Certain > 0.5 Impact virtually certain; emergency response

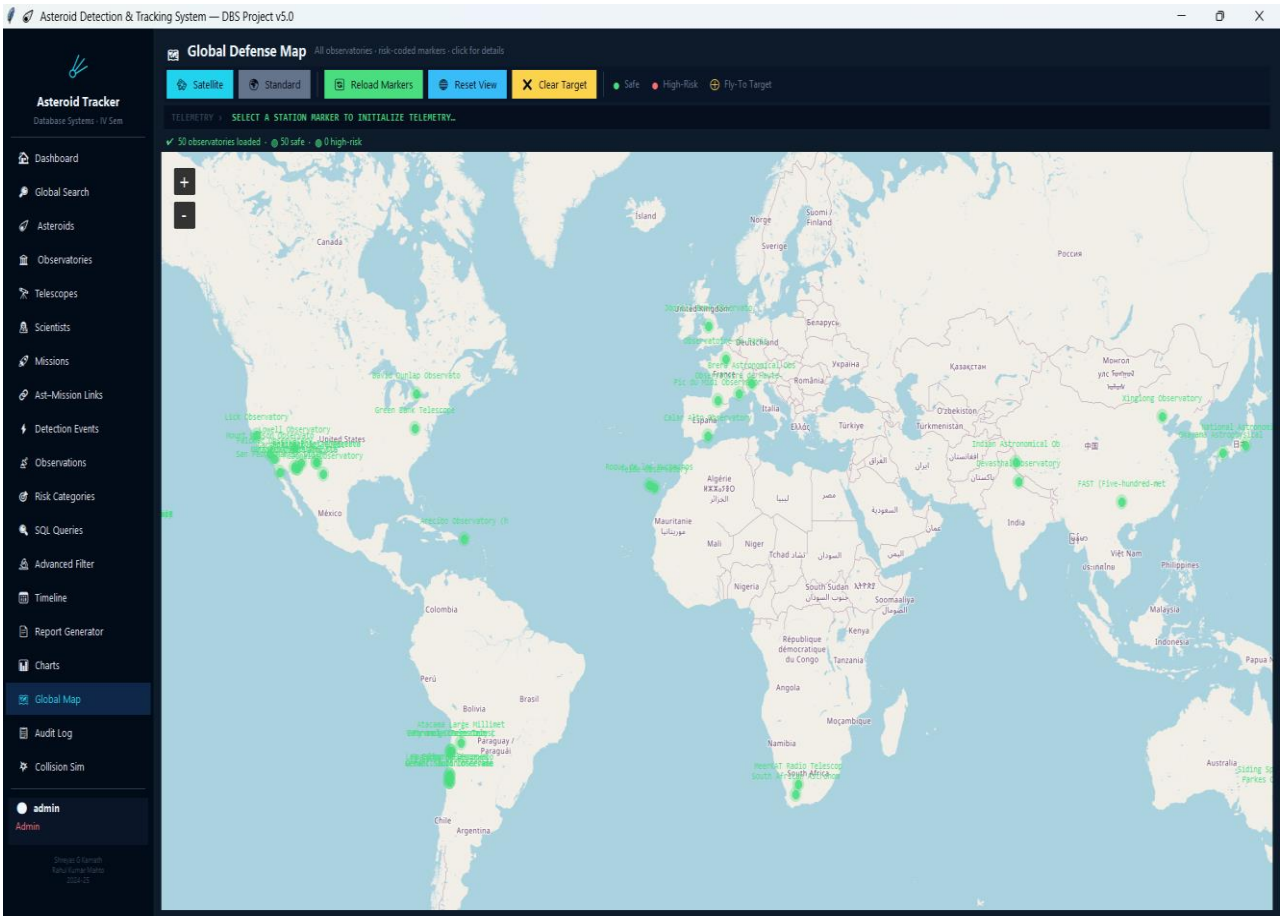

Asteroid Detection & Tracking System — DBS Project v5.0
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Analytical SQL Queries
10 pre-built analytical queries + custom SQL executor
Select Query:
Q1 – High-Risk Asteroids (≥ Moderate)
Run
Export CSV
Custom SQL
Execute Custom SQL
printf('%.6f',de.Probability)
Category_Name
Detection_Date
Name
511 Davida 0.002200 Moderate 2026-03-23 Steven Chesley
1580 Betulia 0.002134 Moderate 2023-07-05 Jon Giorgini
225088 Gonggong 0.001890 Moderate 2024-07-28 Jon Giorgini
16 Psyche 0.001670 Moderate 2025-10-08 Paul Chodas

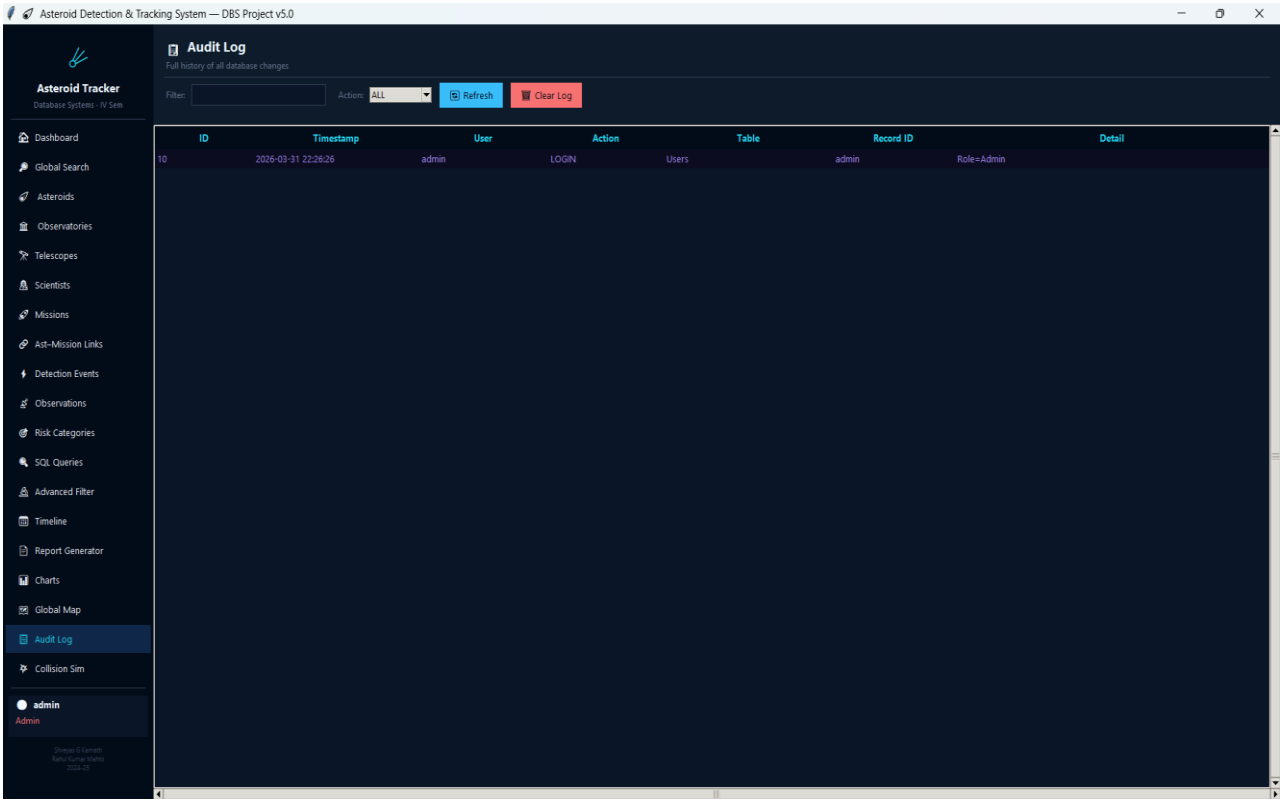

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Advanced Filtering
Combine multiple conditions to drill into detection events
Filter Criteria
Diameter Min (km)
Diameter Max (km)
Risk ID Min (1-10)
Risk ID Max (1-10)
Date From (YYYY-MM-DD)
Date To (YYYY-MM-DD)
Probability Min (0-1)
Probability Max (0-1)
Apply Filters
Reset
Export Results
Asteroid
Diameter
Detection Date
Probability
Risk
Scientist
511 Davida 289.0 2026-03-23 0.002200 Moderate Steven Chesley
1580 Betulia 5.8 2023-07-05 0.002134 Moderate Jon Giorgini
225088 Gonggong 1230.0 2024-07-28 0.001890 Moderate Jon Giorgini
16 Psyche 226.0 2025-10-08 0.001670 Moderate Paul Chodas
4179 Toutatis 2.45 2024-02-14 0.001245 Low-Moderate Lance Benner
3200 Phaethon 5.1 2024-01-19 0.000312 Low-Moderate Michael Brown
33342 1998 WT24 0.41 2025-09-30 0.000234 Low-Moderate Paul Chodas
15760 Albion 108.0 2023-03-29 0.000189 Low-Moderate Richard Binzel
2002 MS4 730.0 2026-05-19 0.000156 Low-Moderate Steven Chesley
243 Ida 31.4 2024-07-25 0.000145 Low Michael Brown
28978 Ixion 710.0 2025-11-03 0.000134 Low Chad Trujillo
1620 Geographos 2.56 2024-10-25 0.000123 Low Francesca DeMeo
162173 Ryugu 0.9 2023-08-09 0.000089 Low Lindley Johnson
704 Interamnia 326.0 2023-05-11 0.000089 Low Lance Benner
4660 Nereus 0.33 2025-05-13 0.000089 Low Vishnu Reddy
2 Pallas 512.0 2025-06-20 0.000078 Low Vishnu Reddy
50000 Quaoar 1086.0 2025-01-21 0.000078 Low David Jewitt
120347 Salacia 870.0 2025-09-30 0.000078 Low Lindley Johnson
25143 Itokawa 0.33 2023-12-18 0.000067 Low Steven Chesley
1566 Icarus 1.4 2023-09-28 0.000067 Low David Jewitt
341843 2008 EV5 0.45 2026-02-24 0.000067 Low Lindley Johnson
84522 2002 TC302 590.0 2026-04-23 0.000067 Low Francesca DeMeo
21 Lutetia 100.0 2025-01-06 0.000056 Low Megan Schwamb
38628 Huya 406.0 2025-08-10 0.000056 Low Scott Sheppard
2100 Ra-Shalom 2.3 2024-05-08 0.000054 Low Chad Trujillo
52 Europa 315.0 2026-01-27 0.000045 Low Alan Harris
136472 Makemake 1430.0 2024-04-04 0.000045 Low Lance Benner

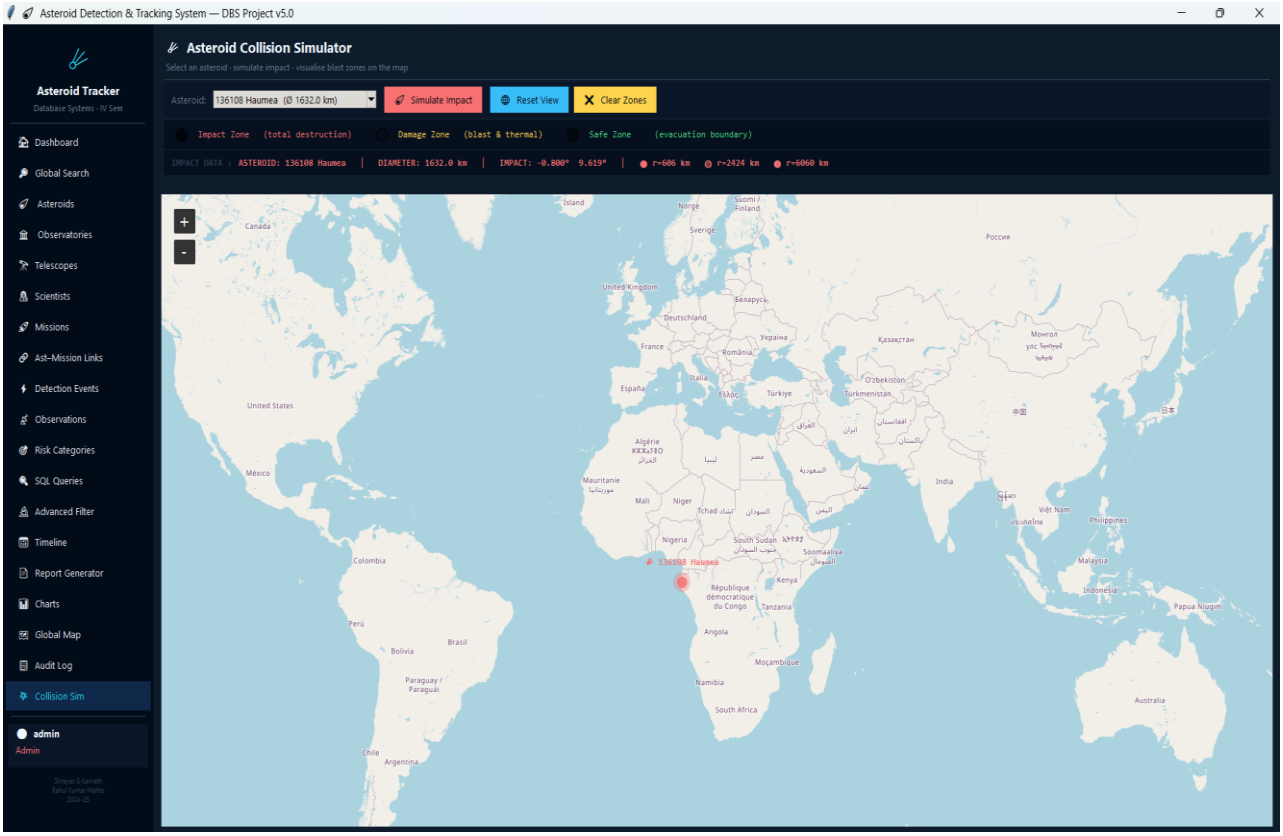

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Event Timeline
Chronological detection & observation events
Asteroid ID:
Show Timeline
Show All
Date
Event Type
Asteroid
Details
Scientist
2023-02-10 Observation 243 Ida RA 04h55m41s Dec +45d11m29s Chad Trujillo
2023-03-15 Observation 99942 Apophis RA 14h52m33s Dec -12d45m22s Amy Mainzer
2023-03-29 Detection 15760 Albion P=0.000189 Risk=Low-Moderate Richard Binzel
2023-04-02 Observation 101955 Bennu RA 05h18m09s Dec +28d19m47s Paul Chodas
2023-04-22 Detection 99942 Apophis P=0.000034 Risk=Low Amy Mainzer
2023-05-11 Detection 704 Interamnia P=0.000089 Risk=Low Lance Benner
2023-05-17 Observation 433 Eros RA 16h46m03s Dec -11d27m58s Chad Trujillo
2023-05-20 Observation 162173 Ryugu RA 09h44m21s Dec -35d01m15s Lindley Johnson
2023-06-05 Observation 951 Gaspra RA 20h33m14s Dec -02d47m36s Megan Schwamb
2023-06-15 Detection 101955 Bennu P=0.000012 Risk=Very Low Paul Chodas
2023-07-05 Detection 1580 Betulia P=0.002134 Risk=Moderate Jon Giorgini
2023-07-08 Observation 433 Eros RA 22h11m55s Dec +15d33m59s Alan Harris
2023-08-09 Detection 162173 Ryugu P=0.000089 Risk=Low Lindley Johnson
2023-08-11 Observation 253 Mathilde RA 04h27m52s Dec -35d18m22s Vishnu Reddy
2023-08-25 Observation 25143 Itokawa RA 03h29m47s Dec -48d22m14s Steven Chesley
2023-09-28 Detection 1566 Icarus P=0.000067 Risk=Low David Jewitt
2023-09-28 Observation 10 Hygiea RA 17h25m59s Dec -08d15m44s Lindley Johnson
2023-10-03 Detection 433 Eros P=0.000005 Risk=Very Low Alan Harris
2023-10-12 Observation 4179 Toutatis RA 17h36m04s Dec +09d08m31s Lance Benner
2023-10-20 Detection 7066 Nessus P=0.000034 Risk=Low Vishnu Reddy
2023-11-16 Detection 136849 1998 CS1 P=0.000012 Risk=Very Low Alan Harris
2023-11-25 Observation 21 Lutetia RA 18h07m33s Dec -06d29m50s Alan Harris
2023-11-30 Observation 216 Kleopatra RA 11h23m18s Dec -20d55m40s Jon Giorgini
2023-12-18 Detection 25143 Itokawa P=0.000067 Risk=Low Steven Chesley
2023-12-19 Observation 511 Davida RA 06h18m31s Dec -44d33m09s Jon Giorgini
2024-01-19 Detection 3200 Phaethon P=0.000312 Risk=Low-Moderate Michael Brown
2024-01-19 Observation 253 Mathilde RA 06h55m42s Dec +32d17m03s David Jewitt
2024-02-09 Detection 136108 Haumea P=0.000003 Risk=Zero Steven Chesley
2024-02-14 Detection 4179 Toutatis P=0.001245 Risk=Low-Moderate Lance Benner
2024-02-18 Observation 2867 Steins RA 14h09m28s Dec +18d04m12s Francesca DeMeo
2024-03-07 Observation 243 Ida RA 20h04m59s Dec -05d44m12s Michael Brown
2024-03-14 Detection 3554 Amun P=0.000018 Risk=Very Low Scott Sheppard
2024-03-29 Observation 25143 Itokawa RA 07h59m21s Dec +22d19m05s Megan Schwamb


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Report Generator
Generate summary reports for asteroids, missions, and risk analysis
Asteroid Summary
Missions Report
Risk Summary
Scientists Report
Export Text
ASTEROID CATALOGUE — DBS PROJECT v4.1
Generated: 2026-03-31 22:26 · Total: 51 asteroids
[35] 136108 Haumea Q=1632.0 km T=284.0 yr
[36] 136472 Makemake Q=1430.0 km T=306.0 yr
[37] 225088 Gonggong Q=1230.0 km T=550.0 yr
[38] 50000 Quaoar Q=1086.0 km T=285.0 yr
[39] 90377 Sedna Q=995.0 km T=11400.0 yr
[13] 1 Ceres Q=946.0 km T=4.61 yr
[49] 2002 AW197 Q=940.0 km T=325.0 yr
[48] 120347 Salacia Q=870.0 km T=284.0 yr
[50] 2002 MS4 Q=730.0 km T=270.0 yr
[41] 28978 Ixion Q=710.0 km T=250.0 yr
[43] 84522 2002 TC302 Q=590.0 km T=410.0 yr
[14] 4 Vesta Q=525.0 km T=3.63 yr
[15] 2 Pallas Q=512.0 km T=4.62 yr
[16] 10 Hygiea Q=434.0 km T=5.59 yr
[40] 38628 Huya Q=406.0 km T=250.0 yr
[21] 704 Interamnia Q=326.0 km T=5.73 yr
[42] 55636 2002 TX300 Q=320.0 km T=283.0 yr
[19] 52 Europa Q=315.0 km T=5.63 yr
[20] 511 Davida Q=289.0 km T=5.66 yr
[18] 65 Cybele Q=237.0 km T=6.08 yr
[17] 16 Psyche Q=226.0 km T=5.0 yr
[47] 54598 Bienor Q=200.0 km T=67.0 yr
[7] 216 Kleopatra Q=135.0 km T=4.32 yr
[44] 15760 Albion Q=108.0 km T=248.0 yr
[12] 21 Lutetia Q=100.0 km T=3.8 yr
[46] 8405 Asbolus Q=84.0 km T=76.0 yr
[45] 7066 Nessus Q=60.0 km T=122.0 yr
[8] 253 Mathilde Q=53.0 km T=4.31 yr
[4] 433 Eros Q=34.4 km T=1.76 yr
[9] 243 Ida Q=31.4 km T=4.84 yr
[10] 951 Gaspra Q=18.2 km T=3.59 yr
[100] ffff Q=10.0 km T=1.0 yr
[22] 1580 Betulia Q=5.8 km T=2.2 yr
[11] 2867 Steins Q=5.3 km T=3.54 yr
[24] 3200 Phaethon Q=5.1 km T=1.43 yr
[28] 1620 Geographos Q=2.56 km T=1.39 yr
[6] 4179 Toutatis Q=2.45 km T=4.03 yr
[26] 2100 Ra-Shalom Q=2.3 km T=0.76 yr
[25] 3554 Amun Q=2.0 km T=0.7 yr
[27] 1862 Apollo Q=1.5 km T=1.81 yr
[23] 1566 Icarus Q=1.4 km T=1.12 yr
[29] 4769 Castalia Q=1.4 km T=0.64 yr
[31] 29075 1950 DA Q=1.3 km T=2.39 yr
[34] 136849 1998 CS1 Q=1.2 km T=2.38 yr
[3] 162173 Ryugu Q=0.9 km T=1.3 yr
[2] 101955 Bennu Q=0.49 km T=1.2 yr
[33] 341843 2008 EV5 Q=0.45 km T=0.96 yr
[32] 33342 1998 WT24 Q=0.41 km T=0.57 yr
[1] 99942 Apophis Q=0.37 km T=0.89 yr
[5] 25143 Itokawa Q=0.33 km T=1.52 yr
[30] 4660 Nereus Q=0.33 km T=1.82 yr


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Data Visualizations
13 interactive chart types · dark-mode · filters
Chart:
Bar – Top 15 Asteroids by Diameter
Year:
Risk≥:
Draw
Save PNG
Top 15 Asteroids by Diameter
Mean
136108 Haumea
136472 Makemake
225088 Gonggong
50000 Quaoar
90377 Sedna
1 Ceres
2002 AW197
120347 Salacia
2002 MS4
28978 Ixion
84522 2002 TC302
4 Vesta
2 Pallas
10 Hygiea
38628 Huya
1632
1430
1230
1086
995
946
940
870
730
710
590
525
512
434
406
0
200
400
600
800
1000
1200
1400
1600
Diameter (km)

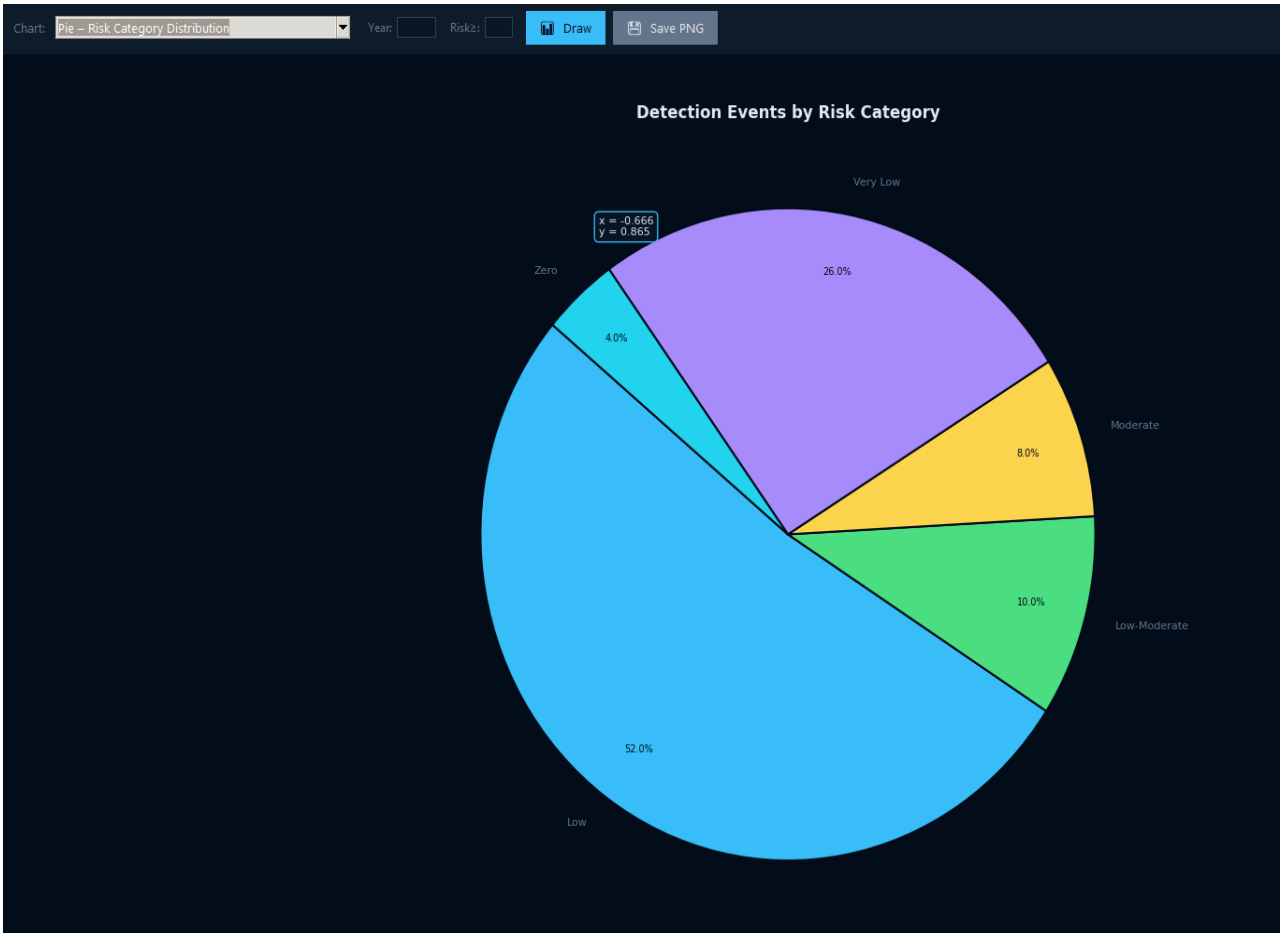

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Global Defense Map
All observatories · risk-coded markers · click for details
Satellite
Standard
Reload Markers
Reset View
Clear Target
Safe
High-Risk
Fly-To Target
TELEMETRY › SELECT A STATION MARKER TO INITIALIZE TELEMETRY…
50 observatories loaded · 50 safe · 0 high-risk


Asteroid Detection & Tracking System — DBS Project v5.0
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Database Systems - IV Sem
Audit Log
Full history of all database changes
Filter:
Action:
ALL
Refresh
Clear Log
ID
Timestamp
User
Action
Table
Record ID
Detail
10
2026-03-31 22:26:26
admin
LOGIN
Users
admin
Role=Admin


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admin
Admin
Asteroid Collision Simulator
Select an asteroid · simulate impact · visualise blast zones on the map
Asteroid:
136108 Haumea (Ø 1632.0 km)
Simulate Impact
Reset View
Clear Zones
Impact Zone (total destruction)
Damage Zone (blast & thermal)
Safe Zone (evacuation boundary)
IMPACT DATA › ASTEROID: 136108 Haumea | DIAMETER: 1632.0 km | IMPACT: -0.800° 9.619° | r=606 km r=2424 km r=6060 km
136108 Haumea

CHARTS:-


Chart:
Horizontal Bar – Scientist Experience
Year:
Risk≥:
Draw
Save PNG
Scientist Experience (Top 15)
Martin Duncan
Richard Binzel
David Jewitt
Alan Harris
Harold Levison
Alberto Cellino
Lindley Johnson
Jon Giorgini
David Rabinowitz
Brett Gladman
Paul Chodas
Matthew Holman
Lance Benner
Michael Brown
Gordon Garradd
0
10
20
30
40
Years


Chart:
Pie – Risk Category Distribution
Year:
Risk≥:
Draw
Save PNG
Detection Events by Risk Category
Very Low
x = -0.666
y = 0.865
Zero
26.0%
4.0%
Moderate
8.0%
10.0%
Low-Moderate
52.0%
Low

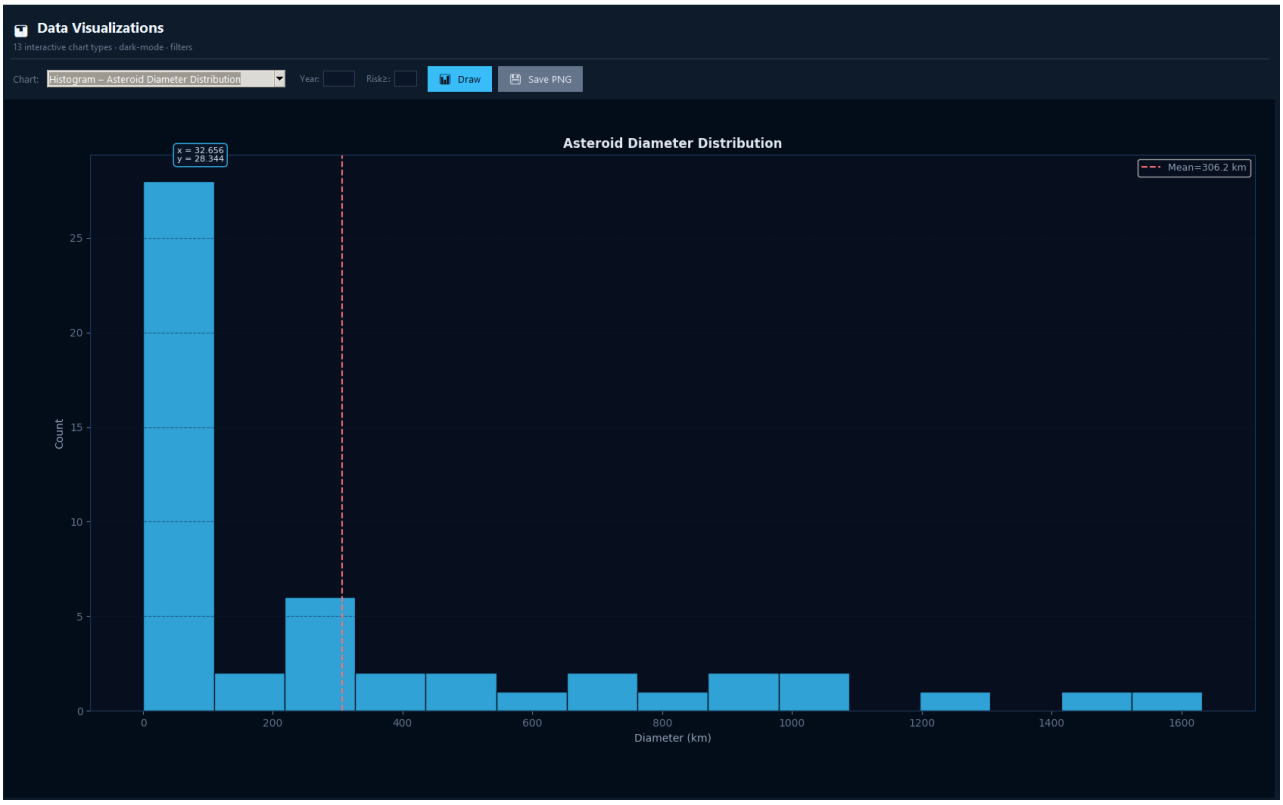
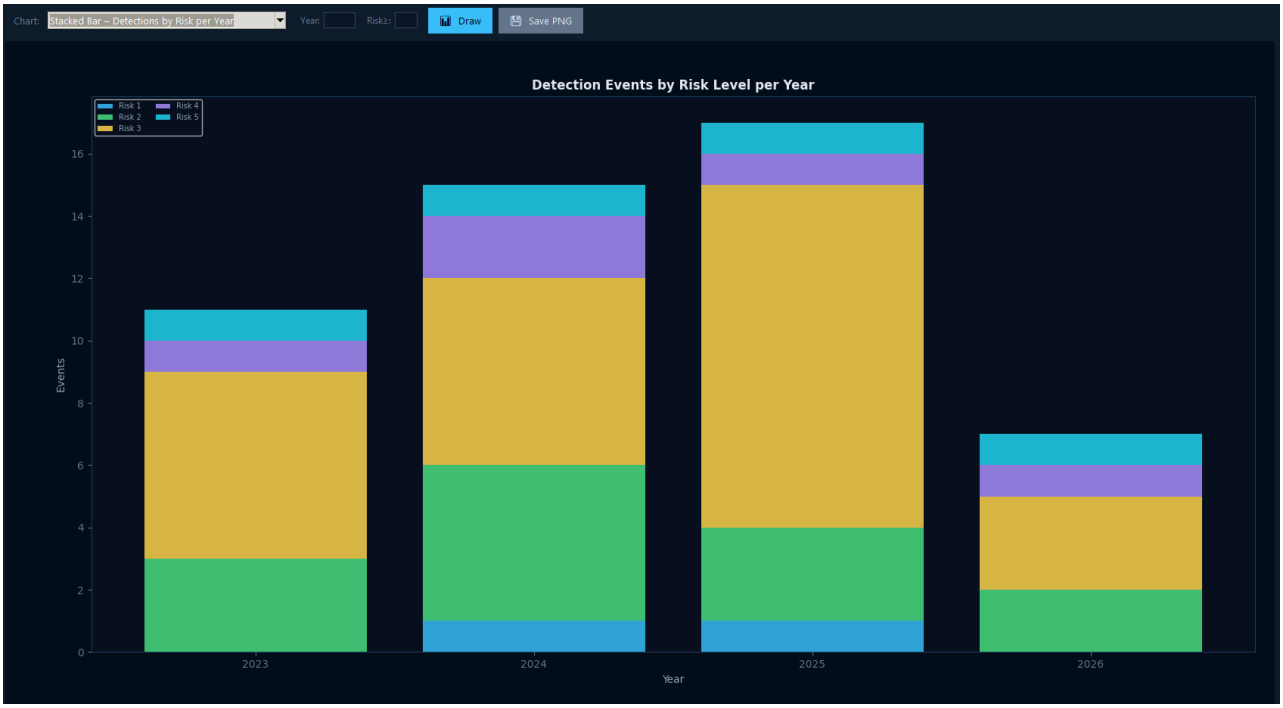
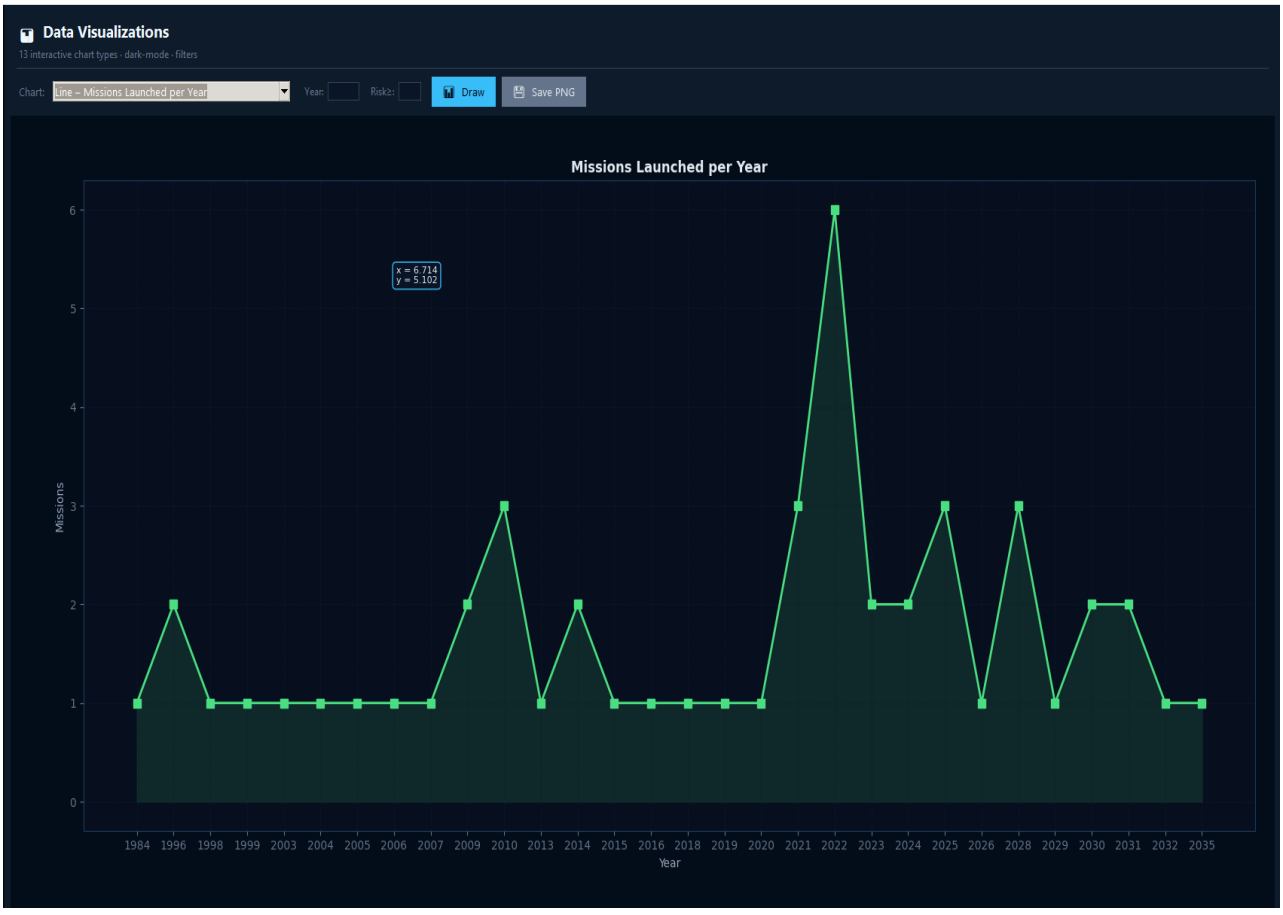
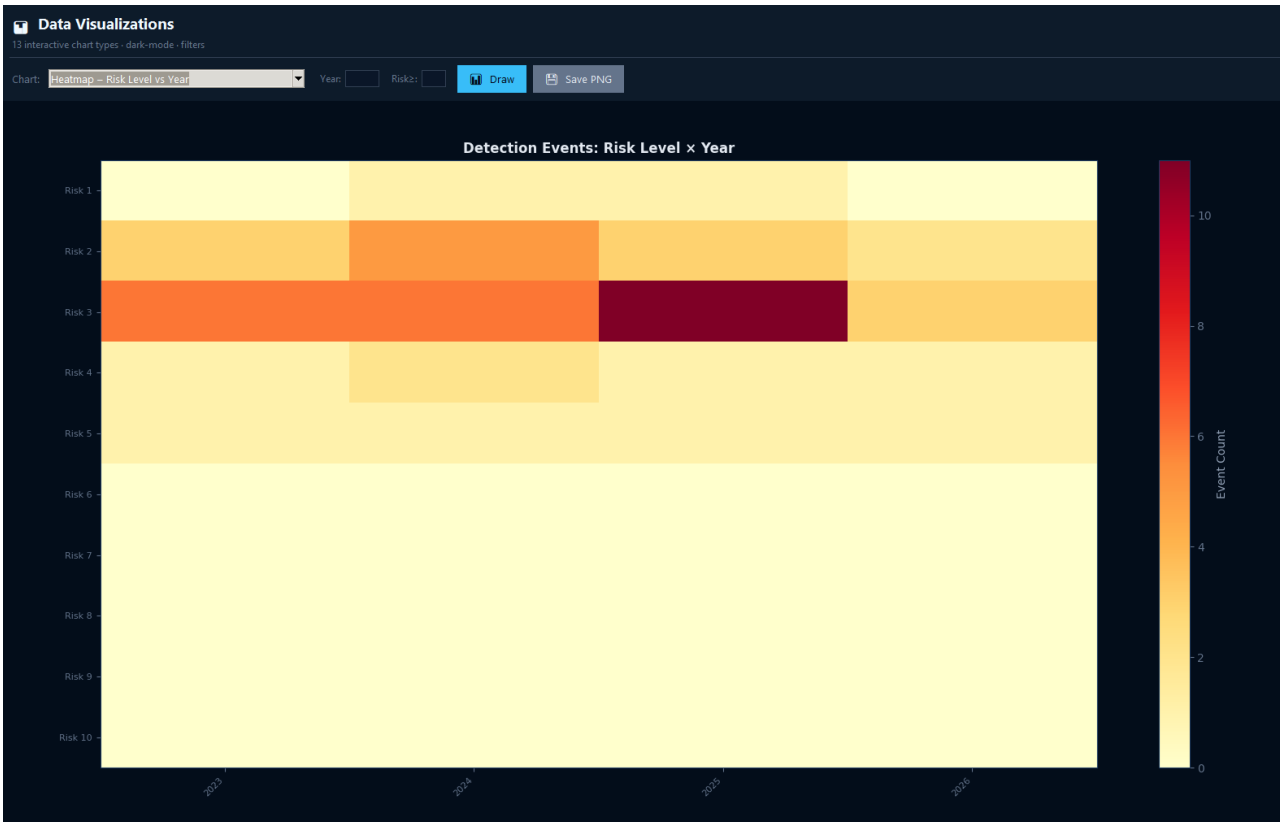

Data Visualizations
13 interactive chart types · dark-mode · filters
Chart:
Histogram – Asteroid Diameter Distribution
Year:
Risk≥:
Draw
Save PNG
Asteroid Diameter Distribution
x = 32.656
y = 28.344
Mean=306.2 km
Count
Diameter (km)
0
5
10
15
20
25
0
200
400
600
800
1000
1200
1400
1600


Chart:
Line – Detection Events Over Time
Year:
Risk≥:
Draw
Save PNG
Detection Events Over Time
x = 0.310
y = 17.204
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Events
2023
2024
2025
2026
Year


Chart:
Stacked Bar – Detections by Risk per Year
Year:
Risk≥:
Draw
Save PNG
Detection Events by Risk Level per Year
Risk 1
Risk 2
Risk 3
Risk 4
Risk 5
16
14
12
10
8
6
4
2
0
Events
2023
2024
2025
2026
Year


Data Visualizations
13 interactive chart types · dark-mode · filters
Chart:
Scatter – Diameter vs Impact Probability
Year:
Risk≥:
Draw
Save PNG
Diameter vs Impact Probability
Mean
x = 114.925
y = 0.002
0.0020
0.0015
0.0010
0.0005
0.0000
Probability
0
200
400
600
800
1000
1200
1400
1600
Diameter (km)
0.00200
0.00175
0.00150
0.00125
0.00100
0.00075
0.00050
0.00025
Probability


Data Visualizations
13 interactive chart types · dark-mode · filters
Chart:
Line – Missions Launched per Year
Year:
Risk≥:
Draw
Save PNG
Missions Launched per Year
x = 6.714
y = 5.102
Missions
0
1
2
3
4
5
6
1984
1996
1998
1999
2003
2004
2005
2006
2007
2009
2010
2013
2014
2015
2016
2018
2019
2020
2021
2022
2023
2024
2025
2026
2028
2029
2030
2031
2032
2035
Year


Data Visualizations
13 interactive chart types · dark-mode · filters
Chart:
Heatmap – Risk Level vs Year
Year:
Risk≥:
Draw
Save PNG
Detection Events: Risk Level × Year
Risk 1
Risk 2
Risk 3
Risk 4
Risk 5
Risk 6
Risk 7
Risk 8
Risk 9
Risk 10
2023
2024
2025
2026
Event Count
0
2
4
6
8
10

CHAPTER 6: CONCLUSION, LIMITATIONS & FUTURE WORK

6.1 Conclusion

The ADTS demonstrates that a well-normalised relational database schema, rigorously designed through ER modelling and formal relational mapping, provides a robust, scalable, and auditable foundation for planetary-defence data management. The 3NF-compliant schema with eleven relations and eight relationship types successfully models the full operational domain of an asteroid tracking facility. CHECK constraints, cascading delete semantics, and SET NULL preservation policies ensure semantic correctness under all CRUD patterns. The Audit_Log delivers a comprehensive, immutable provenance trail essential for post-incident review and liability analysis. The ten analytical SQL queries (Q1–Q10) — spanning multi-table JOIN chains, correlated subqueries, anti-join patterns, and aggregate analytics — demonstrate the expressive power of the relational model for threat intelligence. All eight project objectives (O1–O8) have been met: unified schema, referential integrity, domain validation, M:N linkage, audit logging, risk taxonomy, analytical queries, and comprehensive seed data.

6.2 Limitations

- Static Snapshot Model: The schema stores point-in-time records; orbital elements that evolve continuously due to gravitational perturbations are not captured as time series.
- No Native Triggers: Probability-to-risk mapping and audit logging are application-layer functions. A production system should use native AFTER INSERT / AFTER UPDATE / AFTER DELETE triggers to guarantee integrity regardless of access path.
- SQLite Concurrency: Write-serialisation limits throughput under high-concurrency multi-observatory feeds; a client-server RDBMS (PostgreSQL, MySQL) would be required at scale.
- No Full-Text Search: The Position and Detail columns are unstructured text without full-text indexing.
- Ephemeris Data Excluded: Only the derived Orbital_Period is stored; full orbital element sets (semi-major axis, eccentricity, inclination, etc.) are absent.

6.3 Future Work

6. Database Triggers: Replace application-layer audit functions with native AFTER INSERT / AFTER UPDATE / AFTER DELETE triggers.
7. Stored Procedures / UDFs: Encapsulate probability-to-risk mapping as a user-defined function within the database engine.
8. Migration to PostgreSQL: Leverage partial indexes, JSONB columns, table partitioning for time-series data, and the PostGIS extension for geospatial queries.
9. Temporal Data Modelling: Introduce bitemporal tables (valid time and transaction time) for Detection_Events and Observations to track probability revisions over time.
10. RESTful API Layer: Expose the database via an IVOA-conformant API to enable interoperability with global asteroid tracking networks.
11. Machine Learning Integration: Integrate ML-based anomaly detection to flag statistically unusual probability updates for immediate human review.
12. Distributed Architecture: Architect multi-primary replication using Citus or CockroachDB to support globally distributed writes from multiple observatory sites.
13. GIS Integration: PostGIS spatial queries — e.g., observatories within 1,000 km of a predicted impact trajectory, or ground-track intersection with populated areas.

CHAPTER 7: REFERENCES

[1] E. F. Codd, "A Relational Model of Data for Large Shared Data Banks," Communications of the ACM, vol. 13, no. 6, pp. 377–387, June 1970. DOI: 10.1145/362384.362685

[2] R. Hipp, D. Kennedy, and J. Mistachkin, "SQLite: Small. Fast. Reliable. Choose any three," SQLite Consortium, 2023. [Online]. Available: https://www.sqlite.org

[3] R. Elmasri and S. B. Navathe, Fundamentals of Database Systems, 7th ed. Hoboken, NJ: Pearson Education, 2016, ISBN 978-0-13-397077-7.

[4] NASA Jet Propulsion Laboratory, "Center for Near Earth Object Studies (CNEOS)," 2024. [Online]. Available: https://cneos.jpl.nasa.gov

[5] A. W. Harris and G. D'Abramo, "The Population of Near-Earth Asteroids," Icarus, vol. 257, pp. 302–312, 2015. DOI: 10.1016/j.icarus.2015.05.004

[6] ESA Planetary Defence Office, "Hera Mission: Europe's Planetary Defence Mission," 2024. [Online]. Available: https://www.esa.int/Safety_Security/Hera

[7] International Asteroid Warning Network (IAWN), "Guidelines and Recommendations," United Nations OOSA, Vienna, 2023.

[8] Minor Planet Center (MPC), "MPC Orbit Database," International Astronomical Union, 2024. [Online]. Available: https://minorplanetcenter.net

[9] R. Ramakrishnan and J. Gehrke, Database Management Systems, 3rd ed. New York, NY: McGraw-Hill, 2003, ISBN 978-0-07-246563-1.

[10] C. J. Date, An Introduction to Database Systems, 8th ed. Boston, MA: Addison-Wesley, 2004, ISBN 978-0-32-119784-9.

[11] P. Athanasios et al., "Torino Impact Hazard Scale," Icarus, vol. 132, no. 2, pp. 378–387, 1998.

[12] NASA Office of Inspector General, "NASA's Efforts to Identify Near-Earth Objects and Mitigate Hazards," Report No. IG-14-031, September 2014.